

MS 6015: Multivariate Statistical Methods for Business

Sampling Distributions and Inferences about Mean Vector ¹

¹Reference: JW Chapter 4, section 4 and JW Chapter 5 

Outline

Properties of Multivariate Normal

Sampling Distribution of $\bar{X}$ and S
Central Limit Theorem

Hypothesis Testing

Review of Univariate Hypothesis Testing

Multivariate Hotelling's T^2 Statistic

An Application of One-Sample Hotelling's T-Square

Likelihood Ratio Test

Multivariate Density

- Multivariate Normal density function:

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp - 1/2 \{ (\mathbf{x} - \mu)' \Sigma^{-1} (\mathbf{x} - \mu) \}$$

where $-\infty < x_i < \infty$ for $i = 1, \dots, p$.

- We use the notation $\mathbf{X} \sim N_p(\mu, \Sigma)$ to indicate that the p -dimensional random vector $\mathbf{X}$ has a multivariate normal distribution with mean vector μ and covariance matrix Σ .

Properties of Multivariate Normal Distribution

Let $X \sim N_p(\mu, \Sigma)$

1. Linear combinations of components of X are (multivariate) normal.
2. All sub-sets of the components of X are (multivariate) normal.
3. Zero covariance implies that the corresponding components of X are statistical independent.
4. The conditional distributions of the components of X are (multivariate) normal.

Conditional Distributions

Let $(X_{1(q_1 \times 1)}, X_{2(q_2 \times 1)})$ be distributed as $N_{q_1+q_2}(\mu, \Sigma)$ with

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}; \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

and $|\Sigma| > 0$. The conditional distribution of X_1 given $X_2 = x_2$ is a multivariate normal with mean and covariance matrix

$$\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2) \quad \text{and} \quad \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Bivariate Normal

► Bivariate Normal Distribution

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix} \right)$$

$$\text{then } f(x_1|x_2) \sim N_1 \left(\mu_1 + \frac{\sigma_{12}}{\sigma_{22}}(x_2 - \mu_2), \sigma_{11} - \sigma_{12}\left(\frac{\sigma_{12}}{\sigma_{22}}\right) \right)$$

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► Alternative way to express $f(x_1|x_2)$

$$f(x_1|x_2) \sim N_1 \left(\mu_1 + \rho_{12} \frac{\sqrt{\sigma_{11}}}{\sqrt{\sigma_{22}}}(x_2 - \mu_2), \sigma_{11}(1 - \rho_{12}^2) \right)$$

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► What is this

Goal

- ▶ **Inference:** To make a valid conclusion about the means of a population based on a sample (information about the population).
- ▶ When we have p correlated variables, they must be analyzed jointly
- ▶ Simultaneous analysis yields stronger tests, with better error control.

Sampling Distribution of $\bar{X}$

- ▶ In univariate case we know that $\bar{X}$ is normal with mean μ and variance σ^2/n .
- ▶ The result for the multivariate case is analogous. $\bar{X}$ has a normal distribution with mean vector μ and covariance matrix Σ/n .

Sampling Distribution of S

- ▶ Recall the sample variance, $(n - 1)s^2 = \sum_{j=1}^n (x_j - \bar{x})^2$ is distributed as chi-square with $n - 1$ degrees of freedom
- ▶ Sampling distribution of the sample covariance matrix is called the Wishart Distribution.
- ▶ $(n - 1)S$ is distributed as a Wishart matrix with $n - 1$ degrees of freedom.
- ▶ $\bar{X}$ and S are independent.

- └ Sampling Distribution of $\bar{X}$ and S
- └ Central Limit Theorem

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Likelihood Ratio Test

Central Limit Theorem

- ▶ Let $X_1, X_2, \dots, X_n$ be independent observations from any population with mean μ and finite covariance Σ . Then $\sqrt{n}(\bar{X} - \mu)$ has an approximate $N_p(0, \Sigma)$ distribution.

Goal

- ▶ Define a reasonable distance measure. An estimated mean vector that is too "far away" from the hypothesized mean vector μ_0 gives evidence against the null hypothesis.
- ▶ Construct a likelihood ratio test based on the multivariate normal distribution.

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Univariate Hypothesis Testing

- ▶ Is a given value μ_0 a plausible value for the population mean μ ?
- ▶ We formulate the problem as a hypothesis testing problem. The competing hypotheses are
 $H_0 : \mu = \mu_0$ and $H_1 : \mu \neq \mu_0$

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 $H_0 : \mu = \mu_0$ and $H_1 : \mu \neq \mu_0$
- ▶ Given a sample $X_1, X_2, \dots, X_n$ from a normal population, we compute the test statistic

$$t = \frac{(\bar{X} - \mu_0)}{s/\sqrt{n}}$$

$$\text{where } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \text{ and } s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

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- ▶ If t is "small", then $\bar{X}$ and μ_0 are close and we fail to reject H_0

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$$n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0) > t_{n-1}^2(\alpha/2)$$

Confidence Intervals

- ▶ If we fail to reject H_0 , we conclude that μ_0 is close (in units of standard deviations of $\bar{X}$) to $\bar{X}$, and thus is a plausible value for μ .
- ▶ The set of plausible values for μ is the set of all values that lie in the $100(1 - \alpha)\%$ confidence interval for μ :

$$\left[\bar{X} - t_{(n-1)}(\alpha/2) \frac{s}{\sqrt{n}}, \bar{X} + t_{(n-1)}(\alpha/2) \frac{s}{\sqrt{n}} \right]$$

- ▶ The confidence interval consists of all the values that would not be rejected by the α level test of $H_0 : \mu = \mu_0$.

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Hotelling's T^2 Statistic

- ▶ Consider now the problem of testing whether the $p \times 1$ vector μ_0 is a plausible value for the population mean vector μ .
- ▶ The squared distance

$$T^2 = (\bar{X} - \mu_0)' \left(\frac{S}{n} \right)^{-1} (\bar{X} - \mu_0) = n(\bar{X} - \mu_0)' S^{-1} (\bar{X} - \mu_0)$$

is called the Hotelling T^2 statistic.

- ▶ In the expression above,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i; \quad S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})'$$

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$$T^2 \sim \frac{(n-1)p}{(n-p)} F_{p, n-p}$$

- ▶ We reject the null hypothesis $H_0 : \mu = \mu_0$. for the p -dimensional vector μ at level α when

$$T^2 > \frac{(n-1)p}{(n-p)} F_{p, n-p}$$

where $F_{p, n-p}(\alpha)$ is the upper α percentile of the central F distribution with p and $n - p$ degrees of freedom.

Nature of T^2

- ▶ $(n-1)s^2 \sim \chi_{n-1}^2$
- ▶ $t^2 = \sqrt{n}(\bar{X} - \mu_o)(s^2)^{-1}\sqrt{n}(\bar{X} - \mu_o)$
 - ▶ (Normal) (scaled chi-square /df)(normal)
- ▶ The Wishart distribution generalizes the chi-square distribution.
- ▶ We can write

$$T^2 = \sqrt{n}(\bar{X} - \mu_o)' \left(\frac{\sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})'}{n-1} \right) \sqrt{n}(\bar{X} - \mu_o)$$

- ▶ (Multivariate normal)(Wishart random matrix/df)(Multivariate normal)

Example: Evaluating T^2

Let the data matrix for a random sample of size $n = 3$ from a bivariate normal population be

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- ▶ The sampling distribution of T^2

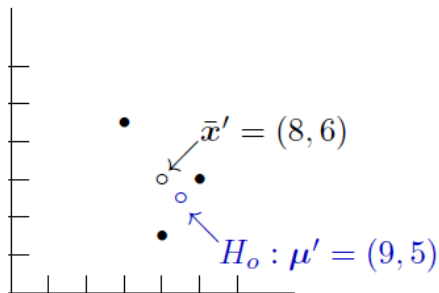
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Graphically



Example: Evaluating T^2 contd.

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Do not reject H_0

Example: Female Sweat Data

- ▶ Perspiration from a sample of 20 healthy females was analyzed. Three variables were measured for each women:
 - X_1 = sweat rate
 - X_2 = sodium content
 - X_3 = potassium content
- ▶ The question is whether $\mu_0 = [4, 50, 10]'$ is plausible for the population mean vector.

Example: Female Sweat Data

- At level $\alpha = 0.1$, we reject the null Hypothesis if

$$\begin{aligned} T^2 = 20(\bar{X} - \mu_0)' S^{-1}(\bar{X} - \mu_0) &> \frac{(n-1)p}{(n-p)} F_{p,n-p}(0.1) \\ &= \frac{19(3)}{17} F_{3,17}(0.1) = 8.18 \end{aligned}$$

- From Data

$$\bar{\bar{x}} = \begin{bmatrix} 4.64 \\ 45.4 \\ 9.96 \end{bmatrix}; \bar{\bar{x}} - \mu_0 = \begin{bmatrix} 4.64 - 4 \\ 45.4 - 50 \\ 9.96 - 10 \end{bmatrix} = \begin{bmatrix} 0.64 \\ -4.6 \\ -0.04 \end{bmatrix}$$

The sample variance covariance matrix is

$$S = \begin{bmatrix} 2.879 & 10.010 & -1.810 \\ 10.010 & 199.788 & -5.640 \\ -1.810 & -5.640 & 3.628 \end{bmatrix}$$

The value of T^2 statistic is 9.74

- ▶ Since $9.74 > 8.18$ we reject H_0 and conclude that μ_0 is not a plausible value for μ at the 10% level.
- ▶ At this point, we do not know which of the three hypothesized mean values is not supported by the data.

Invariance Property of Hotelling's T^2

- ▶ The T^2 statistic is invariant to changes in units of measurements of the form

$$Y_{p \times 1} = C_{p \times p} X_{p \times 1} + d_{p \times 1}$$

with C non-singular. An example of such a transformation is the conversion of temperature measurements from Fahrenheit to Celsius.

- ▶ Given observations $x_1, \dots, x_n$ we find

$$\bar{y} = C\bar{x} + d \text{ and } S_y = CSC'$$

- ▶ $\mu_Y = E(Y) = C\mu + d$ and Hypothesized value
 $\mu_{Y,0} = c\mu_0 + d$
- ▶ $T_y^2 = T_x^2$

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Women's Health Survey: One-Sample Hotelling's T-Square

In 1985, the USDA commissioned a study of women's nutrition. Nutrient intake was measured for a random sample of 737 women aged 25-50 years. Five nutritional components were measured: calcium, iron, protein, vitamin A and vitamin C. In previous analyses of these data, the sample mean vector was calculated. The table below shows the recommended daily intake and the sample means for all the variables:

Variable	Recommended Intake	Mean
Calcium	1000 mg	624.0 mg
Iron	15mg	11.1 mg
Protein	60g	65.8 g
Vitamin A	800 μ g	839.6 μ g
Vitamin C	75 mg	78.9 mg

Questions

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1. One of the questions of interest is whether women meet the federal nutritional intake guidelines. If they fail to meet the guidelines, then we might ask for which nutrients the women fail to meet the guidelines.
2. The hypothesis of interest is that women meet nutritional standards for all nutritional components. This null hypothesis would be rejected if women fail to meet nutritional standards on any one or more of these nutritional variables.

Focus of analysis

- ▶ Univariate: Measuring only a single nutritional component (e.g. Calcium).
- ▶ Multivariate: Measuring multiple (say p) nutritional components (e.g. Calcium, Iron, etc).
 - ▶ Data: $p \times 1$ random vectors

Assumptions

1. Univariate: The data all have a common mean, μ mathematically. This implies that there is a single population of subjects.
2. Multivariate: The data have a common mean vector
$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix}$$
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Assumptions

► Variance

1. Univariate: The data all have a common variance σ^2
2. Multivariate: The data for all subjects have common variance-covariance matrix Σ

► Independence: The subjects are independently sampled.- both univariate and multivariate

► Normality

1. Univariate: The subjects are sampled from a normal distribution
2. Multivariate: The subjects are sampled from a multivariate normal distribution.

Hypothesis

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► Univariate: $H_0 : \mu = \mu_0$ and $H_1 : \mu \neq \mu_0$

► Multivariate: $H_0 : \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \begin{pmatrix} \mu_1^0 \\ \mu_2^0 \\ \vdots \\ \mu_p^0 \end{pmatrix}$

$$H_1 : \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} \neq \begin{pmatrix} \mu_1^0 \\ \mu_2^0 \\ \vdots \\ \mu_p^0 \end{pmatrix}$$

The alternative is that μ_j is not equal to μ_j^0 for at least one j .

Analysis

$$\blacktriangleright S = \begin{pmatrix} 157829.4 & 940.1 & 6075.8 & 102411.1 & 6701.6 \\ 940.1 & 35.8 & 114.1 & 2383.2 & 137.7 \\ 6075.8 & 114.1 & 934.9 & 7330.1 & 477.2 \\ 102411.1 & 2383.2 & 7330.1 & 2668452.4 & 22063.3 \\ 6701.6 & 137.7 & 477.2 & 22063.3 & 5416.3 \end{pmatrix}$$

- ▶ Hotelling's T-square comes out to be: $T^2 = 1758.54$
- ▶ the F - statistic: $349.80 > 3.042 = F_{5,732}(0.01)$

Conclusion

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- ▶ **Message:** For all women between 25 and 50 years old, the average dietary intake does not meet recommended standards. Returning to the table of sample means and recommended dietary intake, it appears that women fail to meet nutritional standards for calcium and iron, and perhaps exceed intakes for protein, vitamin A and vitamin C.

Conclusion

- ▶ We can reject the null hypothesis that the average dietary intake meets recommendations.
- ▶ **Message:** For all women between 25 and 50 years old, the average dietary intake does not meet recommended standards. Returning to the table of sample means and recommended dietary intake, it appears that women fail to meet nutritional standards for calcium and iron, and perhaps exceed intakes for protein, vitamin A and vitamin C.
- ▶ **Question:** For which nutrients do women fall significantly below recommended nutritional intake levels? Or, conversely, for what nutrients do women fall significantly above recommended nutritional intake levels?

Confidence intervals

- ▶ A one-at-a-time 95% confidence interval for calcium is given by the following where values are substituted into the formula and the calculations are worked out as shown below:

$$624.04925 \pm t_{736, 0.025} \sqrt{157829.4/737} = (595.3668, 652.7317)$$

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- ▶ It appears that the average daily intakes of calcium and iron are too low (because the intervals fall below the recommended intakes of these variables), and the average daily intake of protein is too high (because the interval falls above the recommended intake of protein).

Simultaneous Confidence Intervals.

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- ▶ The problem with these one-at-a-time intervals is that they do not control for family-wide error rate.
- ▶ Consequence: We are less than 95% confident that all of the intervals simultaneously cover their respective means.
- ▶ Fix: Calculate a $(1 - \alpha) \times 100\%$ Confidence Ellipse for the population mean vector.

$$n(\bar{X} - \mu)' S^{-1} (\bar{X} - \mu) = \frac{p(n-1)}{(n-p)} F_{p, n-p, \alpha}$$

- ▶ A $(1 - \alpha) \times 100\%$ confidence ellipse yields simultaneous $(1 - \alpha) \times 100\%$ confidence intervals for all linear combinations of the variable means.

$$\begin{aligned} & \text{▶ } 624.04925 \pm \sqrt{\frac{5(737-1)}{737-5} \underbrace{F_{5, 737-5, 0.05}}_{2.21} \sqrt{157829.4/737}} = \\ & (575.27117, 672.82733) \end{aligned}$$

- ▶ Confidence regions are multivariate extensions of univariate confidence intervals.
- ▶ Recall the definition of a $100(1 - \alpha)\%$ CI for a parameter θ : for $X \sim f(x | \theta)$, $\theta \in \Theta$ the interval $(t_1(X), t_2(X))$ is a $100(1 - \alpha)\%$ for θ if

$$Pr[t_1(X) \leq \theta \leq t_2(X)] = 1 - \alpha$$

- ▶ If θ represents a univariate mean μ , a $100(1 - \alpha)\%$ CI for μ is given by

$$\left[\bar{X} - t_{n-1, \frac{\alpha}{2}} \sqrt{s^2/n}, \bar{X} + t_{n-1, \frac{\alpha}{2}} \sqrt{s^2/n} \right]$$

- ▶ Similarly, for $\theta_{p \times 1}$, the region $R(X)$ is a $100(1 - \alpha)\%$ confidence region for θ if

$$Pr[R(X) \text{ will cover the true } \theta] = 1 - \alpha$$

Confidence Regions for Mean

- For the mean vector $\mu_{p \times 1}$, we know that before the sample is selected,

$$P \left[n(\bar{X} - \mu)' S^{-1} (\bar{X} - \mu) \leq \frac{(n-1)p}{(n-p)} F_{p, n-p}(\alpha) \right] = 1 - \alpha$$

meaning that $\bar{X}$ is within statistical distance

$$[(n-1)p F_{p, n-p}(\alpha) / (n-p)]^{1/2}$$

from μ with probability $1 - \alpha$.

- Once a sample is obtained and $\bar{x}$, S are computed, the set of values

$$n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq \frac{(n-1)p}{(n-p)} F_{p, n-p}(\alpha)$$

defines an ellipsoidal region $R(X)$ that is likely to cover μ .

- └ Hypothesis Testing
- └ Likelihood Ratio Test

Outline

Properties of Multivariate Normal

Sampling Distribution of $\bar{X}$ and S
Central Limit Theorem

Hypothesis Testing

Review of Univariate Hypothesis Testing

Multivariate Hotelling's T^2 Statistic

An Application of One-Sample Hotelling's T-Square

Likelihood Ratio Test

Maximum Likelihood Estimation

- ▶ If we consider a $n \times p$ data matrix X with iid rows $X_i \sim N(\mu, \Sigma)$, then the likelihood for an observed data matrix is given by $L(\mu, \Sigma; X)$

$$\begin{aligned} &= \prod_{i=1}^n f(x_i) \\ &= \prod_{i=1}^n (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp \left\{ -\frac{1}{2} (x_i - \mu)' \Sigma^{-1} (x_i - \mu) \right\} \end{aligned}$$

- ▶ The log-likelihood is given by

$$l(\mu, \Sigma; x) = -\frac{np}{2} \log 2\pi - \frac{n}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)' \Sigma^{-1} (x_i - \mu)$$

- ▶ Differentiating the log-likelihood function w.r.t μ and equating to zero we get $\hat{\mu} = \bar{x}$.

Likelihood Ratio Tests and Hotelling's T^2

- ▶ Compare the maximum value of the multivariate normal likelihood function under no restrictions against the restricted maximized value with the mean vector held at μ_0 . The hypothesized value μ_0 will be plausible if it produces a likelihood value almost as large as the unrestricted maximum.
- ▶ To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we construct the ratio

$$LR = \Lambda = \frac{\max_{\Sigma} L(\mu_0, \Sigma)}{\max_{\mu, \Sigma} L(\mu, \Sigma)} = \left(\frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|} \right)^{n/2}$$

where the numerator in the ratio is the likelihood at the MLE of Σ given that $\mu = \mu_0$ and the denominator is the likelihood at the unrestricted MLEs for both μ, Σ .

Wilks' Lambda

- ▶ The equivalent statistic $\Lambda^{2/n} = \frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|}$ is called Wilks' Lambda. If the observed value of this likelihood ratio is too small, then the hypothesis $H_0 : \mu = \mu_0$ is unlikely to be true and therefore rejected.

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$$\Lambda^{2/n} = \frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|} = \left(1 + \frac{T^2}{(n-1)}\right)^{-1}$$

This relation provides another way to compute T^2 that does not require inverting a covariance matrix.

Shape of Confidence Region

- ▶ What can we say about the shape of the confidence region?
- ▶ It is a p -dimensional ellipsoid centered at the sample mean $\bar{X}$
- ▶ Recall that if (λ_i, e_i) are an eigenvalue-eigenvector pair of S , then letting

$$(n-1)pF_{p,n-p}(\alpha)/(n-p) = c^2$$

the i^{th} axis of the confidence ellipse has half length

$$c\sqrt{\frac{\lambda_i}{n}} = \sqrt{(n-1)pF_{p,n-p}(\alpha)/(n-p)}\sqrt{\frac{\lambda_i}{n}}$$

along the e_i direction.

Example: Constructing a confidence ellipse

Consider the microwave oven radiation data in Tables 4.1 and 4.59 (Reference : Johnson and Wichern), where two radiation measurements, x_1 and x_2 , were obtained from $n = 42$ ovens. Here, the x_j denotes the transformed (by Box-Cox) radiation measurements, using a power $\lambda = 0.25$.

Example: Constructing a confidence ellipse

- Sample statistics for those data are:

$$\bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}, \quad S = \begin{bmatrix} 0.014 & 0.012 \\ 0.012 & 0.015 \end{bmatrix}, \quad S^{-1} = \begin{bmatrix} 203.02 & -163.39 \\ -163.39 & 200.23 \end{bmatrix}.$$

- Eigenvalue and eigenvector pairs for S are

$$\lambda_1 = 0.026 \quad e'_1 = [0.704, 0.710]$$

$$\lambda_2 = 0.002 \quad e'_2 = [-0.71, 0.704]$$

95% CR for μ is given by all values μ_1, μ_2 that satisfy:

$$42[0.564 - \mu_1 \quad 0.603 - \mu_2]' \begin{bmatrix} 203.02 & -163.39 \\ -163.39 & 200.23 \end{bmatrix} \begin{bmatrix} 0.564 - \mu_1 \\ 0.603 - \mu_2 \end{bmatrix} \leq 6.62,$$

where

$$\frac{2(41)}{40} F_{2,40}(0.05) = \frac{2(41)}{40} 3.23 = 6.62.$$

Example: Constructing a confidence ellipse

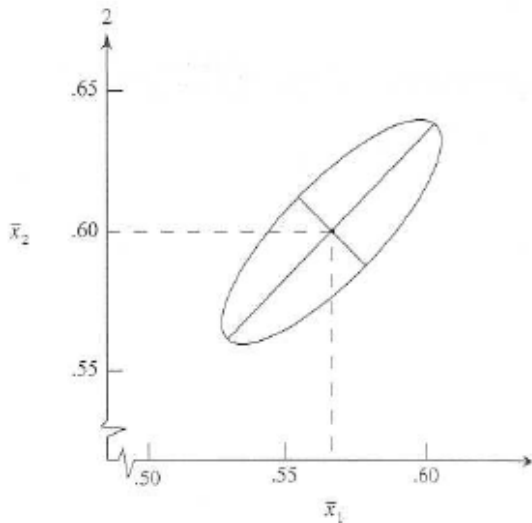
- The joint confidence ellipsoid is centered at $\bar{x} = [0.564 \ 0.603]'$ and the half lengths of the two axes are

$$\sqrt{0.026} \sqrt{\frac{2(41)}{42(40)}} 3.23 = 0.064, \quad \sqrt{0.002} \sqrt{\frac{2(41)}{42(40)}} 3.23 = 0.018.$$

- The axis are in the direction of the two eigenvectors when $\bar{x}$ is taken as the origin.
- The ratio

$$\frac{\sqrt{0.026}}{\sqrt{0.002}} = 3.6$$

indicates that the major axis is 3.6 times longer than the minor axis.



Simultaneous Confidence Intervals

- ▶ Often we are interested in drawing inference about each μ_j
- ▶ One possibility is to construct ordinary confidence intervals

$$\bar{x}_j \pm t_{n-1}\left(\frac{\alpha}{2}\right) \sqrt{\frac{s_{jj}}{n}}$$

for each μ_j .

- ▶ One problem is that the combined set of individual intervals result in a simultaneous confidence level that is less than $1 - \alpha$.
- ▶ There are various ways of constructing a collection of individual confidence intervals so that the joint confidence level for the family of parameters remains at $1 - \alpha$

Simultaneous Confidence Intervals

- Suppose that we have p variables. The population mean of the first variable μ_1 can be written as

$$a'_1\mu = [1 \ 0 \ \dots \ 0]\mu,$$

and in general, $\mu_j = a'_j\mu$ where a'_j is the $p \times 1$ row vector with a one in the j th position and zeros in all other positions.

- Given a sample $x_1, x_2, \dots, x_n$ of p -dimensional vectors, an estimator of μ_j is $a'_j\bar{x}$, with an estimated variance of a'_jSa_j/n .
- Then, an ordinary $(1 - \alpha) \times 100\%$ CI for μ_j can be written as

$$a'_j\bar{x} \pm t_{n-1}(\alpha/2) \sqrt{\frac{a'_jSa_j}{n}}.$$

MS 6015: Multivariate Statistical Methods for Business

Inferences about a Mean Vector ¹

¹Reference: Chapter 5; Johnson and Wichern

Outline

Review of Univariate Hypothesis Testing

Multivariate Hotelling's T^2 Statistic

Likelihood Ratio Test

Confidence Regions

Review of Notation

	Population	Sample Statistics
Mean	μ	$\bar{x}$
Standard Deviation	σ	s
Proportion	p	$\hat{p}$

Univariate Statistics: Tests (Mean / Variance)

Single-sample mean (test $\mu = \mu_0$)

- ▶ t -test (unknown σ):

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- ▶ z -test (known σ):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Univariate Statistics: Tests (Mean / Variance)

Two-sample mean comparison (test $\mu_1 = \mu_2$)

- ▶ Two-sample t -tests:
 - ▶ Pooled t (equal variances)
 - ▶ Welch t (unequal variances)
- ▶ z -test: if population variances are known

Univariate Statistics: Tests (Mean / Variance)

Variance comparison (test $\sigma_1^2 = \sigma_2^2$)

► *F*-test:

$$F = \frac{s_1^2}{s_2^2}$$

Univariate Statistics: Tests (Mean / Variance)

More than two means (3+ groups)

- ▶ ANOVA: tests $\mu_1 = \mu_2 = \dots = \mu_k$
- ▶ Uses an F -ratio:

$$F = \frac{\text{Between-group variability}}{\text{Within-group variability}}$$

Univariate Statistics: Confidence Intervals

- ▶ The $100(1 - \alpha)\%$ confidence interval for p is

$$\left[\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \right]$$

- ▶ The $100(1 - \alpha)\%$ confidence interval for μ is

$$\left[\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}} \right]$$

Univariate Statistics: Confidence Intervals

Univariate Statistics: t -distribution

- ▶ When the population standard deviation is unknown, it must be replaced by the sample statistic
- ▶ This yields the summary statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

- ▶ If a random sample is taken from a normal population, then the statistic has a t distribution with d.f. = $n - 1$
- ▶ As sample size increases, the t -distribution approaches the z -distribution

Univariate Statistics: Confidence Intervals

Univariate Statistics: Confidence Interval for μ

- ▶ The $100(1 - \alpha)\%$ confidence interval for μ is

$$\left[\bar{x} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \right]$$

- ▶ The value of t depends on the level of confidence and $n - 1$ degrees of freedom.

Parts of a Hypothesis Test

1. State Null and alternative Hypothesis
2. Collect Data
3. Calculate the Test Statistic
4. Compute p -value
5. Interpret the results

Recall- Univariate test for mean

- ▶ Is a given value μ_0 a plausible value for the population mean μ ?
- ▶ We formulate the problem as a hypothesis testing problem. The competing hypotheses are

$$H_0 : \mu = \mu_0 \text{ and } H_1 : \mu \neq \mu_0$$

- ▶ Given a sample $x_1, x_2, \dots, x_n$ from a normal population, we compute the test statistic

$$t = \frac{(\bar{x} - \mu_0)}{s/\sqrt{n}}$$

where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \text{ and } s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- ▶ If t is "small", then $\bar{x}$ and μ_0 are close and we fail to reject H_0

Univariate t - statistic

- ▶ When H_0 is true, the statistic t has a student t distribution with $n - 1$ degrees of freedom.
- ▶ We reject the null hypothesis at level α when $|t| > t_{(n-1)}(\alpha/2)$

Univariate t - statistic

- ▶ When H_0 is true, the statistic t has a student t distribution with $n - 1$ degrees of freedom.
- ▶ We reject the null hypothesis at level α when $|t| > t_{(n-1)}(\alpha/2)$
- ▶ Notice that rejecting H_0 when t is large is equivalent to rejecting it when the squared standardized distance

$$t^2 = \frac{(\bar{X} - \mu_0)^2}{s^2/n} = n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0)$$

is large.

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- ▶ We reject H_0 when

$$n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0) > t_{n-1}^2(\alpha/2)$$

Confidence Intervals

- ▶ If we fail to reject H_0 , we conclude that μ_0 is close (in units of standard deviations of $\bar{x}$) to $\bar{x}$, and thus is a plausible value for μ .
- ▶ The set of plausible values for μ is the set of all values that lie in the $100(1 - \alpha)\%$ confidence interval for μ :

$$\left[\bar{x} - t_{(n-1)}(\alpha/2) \frac{s}{\sqrt{n}}, \bar{x} + t_{(n-1)}(\alpha/2) \frac{s}{\sqrt{n}} \right]$$

- ▶ The confidence interval consists of all the values that would not be rejected by the α level test of $H_0 : \mu = \mu_0$.

From one-sample t -test to multivariate testing

Scalar (univariate)

$$H_0 : \mu = \mu_0, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

From one-sample t -test to multivariate testing

Scalar (univariate)

$$H_0 : \mu = \mu_0, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Vector (multivariate)

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0, \quad \bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$$

Recap: one-sample t -test (scalar)

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Recap: one-sample t -test (scalar)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- ▶ Measures **signal**: $\bar{x} - \mu_0$
- ▶ Scales by **noise**: $s/\sqrt{n}$

Multivariate setup

Let $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$ be i.i.d. observations.

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Sample covariance matrix:

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

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- ▶ $\mathbf{S}$ generalizes variance s^2
- ▶ Captures **variability and correlations** across variables

Hotelling's T^2 : multivariate analogue of the t -test

Null hypothesis:

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$$

Hotelling's T^2 : multivariate analogue of the t -test

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Test statistic:

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Interpretation:

- ▶ $(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$ is the **mean-difference vector** (signal)
- ▶ $\mathbf{S}^{-1}$ downweights directions with high variability (noise)
- ▶ T^2 is a **Mahalanobis distance** between $\bar{\mathbf{x}}$ and $\boldsymbol{\mu}_0$

Decision rule (using an F distribution)

- ▶ If the observed T^2 value is "large" we reject $H_0 : \mu = \mu_0$.
- ▶ To decide how large is large, we need the sampling distribution of T^2 when the hypothesized mean vector is correct: If data are approximately multivariate normal, then:

$$\frac{n-p}{p(n-1)} T^2 \sim F_{p, n-p}$$

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- ▶ Works for $n > p$ (so that $\mathbf{S}$ is invertible)
- ▶ When $p = 1$, this reduces to the usual one-sample t -test logic

Example: Evaluating T^2

Let the data matrix for a random sample of size $n = 3$ from a bivariate normal population be

$$X = \begin{bmatrix} 6 & 9 \\ 10 & 6 \\ 8 & 3 \end{bmatrix}$$

- ▶ The observed T^2 for $\mu'_0 = [9, 5]$ is $\frac{7}{9}$
- ▶ The sampling distribution of T^2 is the distribution of $4F_{2,1}$ random variable.

Example: Female Sweat Data

- ▶ Perspiration from a sample of 20 healthy females was analyzed. Three variables were measured for each women:
 - X_1 = sweat rate
 - X_2 = sodium content
 - X_3 = potassium content
- ▶ The question is whether $\mu_0 = [4, 50, 10]'$ is plausible for the population mean vector.

Example: Female Sweat Data

- ▶ At level $\alpha = 0.1$, we reject the null Hypothesis if

$$\begin{aligned} T^2 = 20(\bar{X} - \mu_0)' S^{-1}(\bar{X} - \mu_0) &> \frac{(n-1)p}{(n-p)} F_{p,n-p}(0.1) \\ &= \frac{19(3)}{17} F_{3,17}(0.1) = 8.18 \end{aligned}$$

- ▶ From Data

$$\bar{\mathbf{x}} = \begin{bmatrix} 4.64 \\ 45.4 \\ 9.96 \end{bmatrix}; \bar{\mathbf{x}} - \mu_0 = \begin{bmatrix} 4.64 - 4 \\ 45.4 - 50 \\ 9.96 - 10 \end{bmatrix} = \begin{bmatrix} 0.64 \\ -4.6 \\ -0.04 \end{bmatrix}$$

The sample variance covariance matrix is

$$S = \begin{bmatrix} 2.879 & 10.010 & -1.810 \\ 10.010 & 199.788 & -5.640 \\ -1.810 & -5.640 & 3.628 \end{bmatrix}$$

The value of T^2 statistic is 9.74

- ▶ Since $9.74 > 8.18$ we reject H_0 and conclude that μ_0 is not a plausible value for μ at the 10% level.
- ▶ At this point, we do not know which of the three hypothesized mean values is not supported by the data.

Invariance Property of Hotelling's T^2

- ▶ The T^2 statistic is invariant to changes in units of measurements of the form

$$Y_{p \times 1} = C_{p \times p} X_{p \times 1} + d_{p \times 1}$$

with C non-singular. An example of such a transformation is the conversion of temperature measurements from Fahrenheit to Celsius.

- ▶ Given observations $x_1, \dots, x_n$ we find

$$\bar{y} = C\bar{x} + d \text{ and } S_y = CSC'$$

- ▶ $\mu_Y = E(Y) = C\mu + d$ and Hypothesized value
 $\mu_{Y,0} = c\mu_0 + d$
- ▶ $T_y^2 = T_x^2$

Likelihood Ratio Tests and Hotelling's T^2

- ▶ Compare the maximum value of the multivariate normal likelihood function under no restrictions against the restricted maximized value with the mean vector held at μ_0 . The hypothesized value μ_0 will be plausible if it produces a likelihood value almost as large as the unrestricted maximum.
- ▶ To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we construct the ratio

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where the numerator in the ratio is the likelihood at the MLE of Σ given that $\mu = \mu_0$ and the denominator is the likelihood at the unrestricted MLEs for both μ, Σ .

Wilks' Lambda

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$$\Lambda^{2/n} = \frac{|\hat{\Sigma}|}{|\hat{\Sigma}_0|} = \left(1 + \frac{T^2}{(n-1)}\right)^{-1}$$

This relation provides another way to compute T^2 that does not require inverting a covariance matrix.

ANOVA vs Hotelling T^2 (Core Idea)

ANOVA	Hotelling T^2
Many populations, one variable	One population, many variables
Compare same measurement across groups	Compare several characteristics of one group
Example: average return of Value, Growth, Index funds	Example: one portfolio's return, volatility, liquidity
$H_0 : \mu_1 = \mu_2 = \mu_3$	$H_0 : \mu = \mu_0$
Tests differences between groups	Tests whether a mean vector matches a target

Remember

ANOVA → compare **groups**Hotelling T^2 → compare **a profile (vector)**

Confidence Regions

- ▶ Confidence regions are multivariate extensions of univariate confidence intervals.
- ▶ Recall the definition of a $100(1 - \alpha)\%$ CI for a parameter θ : for $X \sim f(x | \theta), \theta \in \Theta$ the interval $(t_1(X), t_2(X))$ is a $100(1 - \alpha)\%$ for θ if

$$Pr[t_1(X) \leq \theta \leq t_2(X)] = 1 - \alpha$$

- ▶ If θ represents a univariate mean μ , a $100(1 - \alpha)\%$ CI for μ is given by

$$\left[\bar{X} - t_{n-1, \frac{\alpha}{2}} \sqrt{s^2/n}, \bar{X} + t_{n-1, \frac{\alpha}{2}} \sqrt{s^2/n} \right]$$

- ▶ Similarly, for $\theta_{p \times 1}$, the region $R(X)$ is a $100(1 - \alpha)\%$ confidence region for θ if

$$Pr[R(X) \text{ will cover the true } \theta] = 1 - \alpha$$

Confidence Regions for Mean

- For the mean vector $\mu_{p \times 1}$, we know that before the sample is selected,

$$P \left[n(\bar{X} - \mu)' S^{-1} (\bar{X} - \mu) \leq \frac{(n-1)p}{(n-p)} F_{p, n-p}(\alpha) \right] = 1 - \alpha$$

meaning that $\bar{X}$ is within statistical distance

$$[(n-1)p F_{p, n-p}(\alpha) / (n-p)]^{1/2}$$

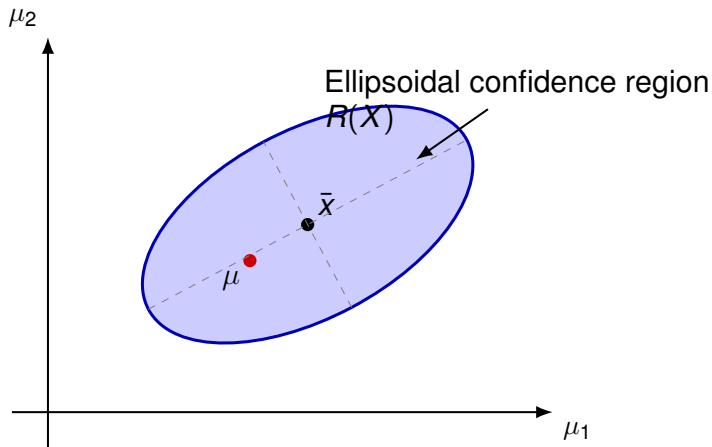
from μ with probability $1 - \alpha$.

- Once a sample is obtained and $\bar{x}$, S are computed, the set of values

$$n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq \frac{(n-1)p}{(n-p)} F_{p, n-p}(\alpha)$$

defines an ellipsoidal region $R(X)$ that is likely to cover μ .

Confidence Region for the Mean Vector



Shape of Confidence Region

- ▶ What can we say about the shape of the confidence region?
- ▶ It is a p -dimensional ellipsoid centered at the sample mean $\bar{X}$
- ▶ Recall that if (λ_i, e_i) are an eigenvalue-eigenvector pair of S , then letting

$$(n-1)pF_{p,n-p}(\alpha)/(n-p) = c^2$$

the i^{th} axis of the confidence ellipse has half length

$$c\sqrt{\frac{\lambda_i}{n}} = \sqrt{(n-1)pF_{p,n-p}(\alpha)/(n-p)}\sqrt{\frac{\lambda_i}{n}}$$

along the e_i direction.

Example: Constructing a confidence ellipse

Consider the microwave oven radiation data in Tables 4.1 and 4.59 (Reference : Johnson and Wichern), where two radiation measurements, x_1 and x_2 , were obtained from $n = 42$ ovens. Here, the x_j denotes the transformed (by Box-Cox) radiation measurements, using a power $\lambda = 0.25$.

Data and Objective

We observe $n = 42$ bivariate observations $x_i = (x_{1i}, x_{2i})'$ (so $p = 2$).

Goal: Construct a $(1 - \alpha)$ confidence region for the (unknown) mean vector μ .

Confidence region:

$$R(X) = \{ \mu : n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq c_\alpha \}, \quad c_\alpha = \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha).$$

For $p = 2$, $R(X)$ is an **ellipse** in the (μ_1, μ_2) -plane, centered at $\bar{x}$.

Step 1: Compute $\bar{x}$ and S from the sample

Given $x_1, \dots, x_n$:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad S = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})'.$$

For your data ($n = 42$), the numerical results are:

$$\bar{x} = \begin{bmatrix} 0.5642575 \\ 0.6029812 \end{bmatrix}$$

$$S = \begin{bmatrix} 0.01435023 & 0.01171547 \\ 0.01171547 & 0.01454530 \end{bmatrix}$$

(Units are whatever units X_1 and X_2 are measured in.)

Step 2: Set the cutoff c_α and rewrite as an ellipse

Choose $\alpha = 0.05$ (95% region). With $p = 2$ and $n = 42$:

$$c_\alpha = \frac{(n-1)p}{n-p} F_{2,40}(0.05).$$

Numerically:

$$F_{2,40}(0.05) \approx 3.2317, \quad \boxed{c_\alpha \approx 6.6250}.$$

So the confidence region is

$$n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq 6.6250 \iff (\mu - \bar{x})' S^{-1} (\mu - \bar{x}) \leq \frac{6.6250}{42}.$$

Define

$$\boxed{k = \frac{c_\alpha}{n} \approx 0.157739.}$$

Geometrically, this is an ellipse centered at $\bar{x}$, with shape determined by S .

Why the Confidence Region is an Ellipse

We obtained the confidence region

$$(\mu - \bar{x})' S^{-1} (\mu - \bar{x}) \leq k, \quad k = \frac{c_\alpha}{n} = 0.157739.$$

The shape of the region is determined by the covariance matrix

$$S = \begin{bmatrix} 0.01435023 & 0.01171547 \\ 0.01171547 & 0.01454530 \end{bmatrix}.$$

Key idea: Diagonalize S using eigenvalues and eigenvectors.

$$S = P \Lambda P'$$

Eigenvalues:

$$\lambda_1 = 0.02619, \quad \lambda_2 = 0.00271$$

- ▶ Eigenvectors give the **direction** of the ellipse
- ▶ Eigenvalues determine the **length of the axes**

Eigen-Geometry of the 95% Confidence Ellipse for μ

For your data: $n = 42$, $p = 2$, $\alpha = 0.05$.

$$\bar{x} = \begin{bmatrix} 0.5642575 \\ 0.6029812 \end{bmatrix}, \quad S = \begin{bmatrix} 0.01435023 & 0.01171547 \\ 0.01171547 & 0.01454530 \end{bmatrix}, \quad k = \frac{c_\alpha}{n} = 0.157739.$$

Diagonalize $S = P\Lambda P'$ (principal directions):

$$\lambda_1 = 0.02616364, \quad v_1 = \begin{bmatrix} 0.704157 \\ 0.710044 \end{bmatrix} \quad (\text{major direction})$$

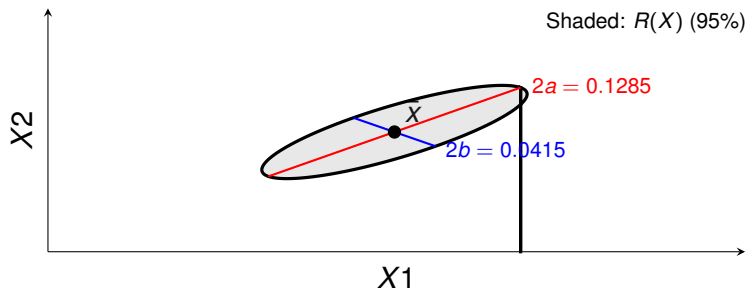
$$\lambda_2 = 0.00273189, \quad v_2 = \begin{bmatrix} -0.710044 \\ 0.704157 \end{bmatrix} \quad (\text{minor direction})$$

Axis lengths of the ellipse $(\mu - \bar{x})' S^{-1} (\mu - \bar{x}) \leq k$:

$$\text{semi-major } a = \sqrt{k\lambda_1} = 0.064242 \quad \Rightarrow \quad \text{major axis length } 2a = 0.128484$$

$$\text{semi-minor } b = \sqrt{k\lambda_2} = 0.020759 \quad \Rightarrow \quad \text{minor axis length } 2b = 0.041518$$

Example: Constructing a confidence ellipse



Simultaneous Confidence Intervals

- ▶ Often we are interested in drawing inference about each μ_j
- ▶ One possibility is to construct ordinary confidence intervals

$$\bar{x}_j \pm t_{n-1}\left(\frac{\alpha}{2}\right) \sqrt{\frac{s_{jj}}{n}}$$

for each μ_j .

- ▶ One problem is that the combined set of individual intervals result in a simultaneous confidence level that is less than $1 - \alpha$.
- ▶ There are various ways of constructing a collection of individual confidence intervals so that the joint confidence level for the family of parameters remains at $1 - \alpha$

Confidence Interval for a Single Component of a Mean Vector

Suppose we have p variables and a population mean vector

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}.$$

The mean of the j^{th} variable can be written using a selector vector

$$\mathbf{a}'_j = [0 \ \cdots \ 1 \ \cdots \ 0],$$

where the 1 is in the j^{th} position. Then

$$\mu_j = \mathbf{a}'_j \boldsymbol{\mu}.$$

Simultaneous Confidence Intervals

Given a sample $x_1, x_2, \dots, x_n$ of p -dimensional observations,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

is the estimator of μ , and an estimator of μ_j is

$$\hat{\mu}_j = a_j' \bar{x}.$$

Its estimated variance is

$$\text{Var}(\hat{\mu}_j) = \frac{a_j' S a_j}{n},$$

where S is the sample covariance matrix.

Ordinary $(1 - \alpha)100\%$ confidence interval for μ_j :

$$a_j' \bar{x} \pm t_{n-1} \left(\frac{\alpha}{2} \right) \sqrt{\frac{a_j' S a_j}{n}}.$$

Simultaneous Confidence Intervals (Hotelling T^2)

Result: Let $X_1, X_2, \dots, X_n$ be a random sample from a p -variate normal population

$$N_p(\mu, \Sigma),$$

where Σ is positive definite.

Then, *simultaneously for all linear combinations $a'\mu$* , the interval

$$\left(a'\bar{X} - \sqrt{\frac{p(n-1)}{n(n-p)}} F_{p,n-p}(\alpha) a'Sa, a'\bar{X} + \sqrt{\frac{p(n-1)}{n(n-p)}} F_{p,n-p}(\alpha) a'Sa \right)$$

contains $a'\mu$ with probability $1 - \alpha$.

Interpretation

- ▶ a = any vector of constants (selects any linear combination of the variables)
- ▶ Gives **simultaneous** confidence intervals for all components of μ
- ▶ This is the interval form of the Hotelling T^2 confidence region

Componentwise T^2 Simultaneous Confidence Intervals

The simultaneous intervals from the Result are called T^2 -intervals, since their coverage probability is determined by the distribution of Hotelling's T^2 statistic.

Choose the vectors

$$a'_1 = [1, 0, \dots, 0], a'_2 = [0, 1, \dots, 0], \dots, a'_p = [0, 0, \dots, 1].$$

These select each component of the mean vector μ .

Then the following intervals hold **simultaneously** with confidence $1 - \alpha$:

$$\bar{x}_j - \sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{jj}}{n}} \leq \mu_j \leq \bar{x}_j + \sqrt{\frac{p(n-1)}{n-p} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{jj}}{n}}, \quad j = 1, \dots, p.$$

Important

All intervals are valid *at the same time*.

Problem with Simultaneous Confidence Intervals

The problem with the simultaneous confidence intervals is that if we are not interested in all possible linear combinations of variables or anything other than the means by themselves, then the simultaneous confidence interval may be too wide, and hence, too conservative. As an alternative to the simultaneous confidence intervals, we can use the Bonferroni intervals.

Bonferroni Confidence Intervals

For the j^{th} variable ($j = 1, \dots, p$),

$$\bar{x}_j \pm t_{n-1, \alpha/(2p)} \sqrt{\frac{s_j^2}{n}}$$

where:

- ▶ $\bar{x}_j$ = sample mean of variable j
- ▶ s_j^2 = sample variance of variable j
- ▶ n = sample size
- ▶ $t_{n-1, \alpha/(2p)}$ = upper $\alpha/(2p)$ quantile of the t distribution

One-at-a-Time vs. Bonferroni vs. Hotelling's T^2

Goal: Confidence statements about a multivariate mean vector $\mu = (\mu_1, \dots, \mu_p)^\top$ based on a sample of size n .

1) One-at-a-Time (Marginal) CIs

For each j :

$$\bar{x}_j \pm t_{n-1, \alpha/2} \sqrt{\frac{s_j^2}{n}}$$

Guarantee: $P(\mu_j \in I_j) = 1 - \alpha$ (for each j).

But: $P(\mu_1 \in I_1, \dots, \mu_p \in I_p)$ is typically $< 1 - \alpha$.

2) Bonferroni Simultaneous CIs

For each j :

$$\bar{x}_j \pm t_{n-1, \alpha/(2p)} \sqrt{\frac{s_j^2}{n}}$$

Guarantee: $P(\mu_1 \in I_1, \dots, \mu_p \in I_p) \geq 1 - \alpha$.

Tradeoff: Conservative (intervals wider), but simple and general.

3) Hotelling's T^2 (Ellipsoidal Region $\Rightarrow$ Simultaneous CIs)

Assuming multivariate normality, a $(1 - \alpha)$ confidence region for μ is

$$n(\bar{\mathbf{x}} - \mu)^\top S^{-1}(\bar{\mathbf{x}} - \mu) \leq \frac{p(n-1)}{n-p} F_{p, n-p}(1 - \alpha).$$

Interpretation: An ellipsoid in $\mathbb{R}^p$ giving *joint* coverage $1 - \alpha$. Componentwise simultaneous CIs can be derived from this region (often less conservative than Bonferroni).

Univariate vs Multivariate Inference

Univariate (one variable)	Multivariate (several variables)
Single Population	
One-sample t -test Test: $H_0 : \mu = \mu_0$	One-sample Hotelling's T^2 Test: $H_0 : \mu = \mu_0$
Comparing Two Populations	
Two-sample/Paired t -test Test: $H_0 : \mu_1 = \mu_2$	Two-sample/Paired Hotelling's T^2 Test: $H_0 : \mu_1 = \mu_2$
Comparing Several Populations	
ANOVA Test: $H_0 : \mu_1 = \mu_2 = \dots = \mu_g$	MANOVA Test: $H_0 : \mu_1 = \mu_2 = \dots = \mu_g$

Key Idea:

- ▶ Univariate methods analyze **one response variable**.
- ▶ Multivariate methods analyze a **vector of correlated responses**.
- ▶ Hotelling's T^2 is the multivariate analogue of the t -test.

Summary

1. Hotelling's T^2 statistic to test mean vector of one sample
2. Constructing confidence regions
3. Simultaneous confidence intervals
4. Bonferroni confidence intervals

MS 6015: Multivariate Statistical Methods for Business

Comparison of Several Multivariate Means ¹

¹Reference: JW Chapter 6, Section 1-3

From Last Class

Hypothesis Testing(Inferences about a Mean vector)

- ▶ Review of Univariate Hypothesis Testing
- ▶ Multivariate Hotelling's T^2 Statistic
- ▶ Likelihood Ratio Test
- ▶ Confidence Regions and Simultaneous Confidence intervals

Outline

Two Independent Samples: Univariate Case
Difference between Two Means

Two Independent Samples: Multivariate

Paired Comparisons

Paired Comparisons-Univariate

Paired Comparisons-Multivariate

- └ Two Independent Samples: Univariate Case
 - └ Difference between Two Means

Outline

Two Independent Samples: Univariate Case

Difference between Two Means

Two Independent Samples: Multivariate

Paired Comparisons

Paired Comparisons-Univariate

Paired Comparisons-Multivariate

Example

- ▶ A fitness chain is considering licensing a proprietary diet at a cost of \$200,000. Is it more effective than the conventional free government recommended food pyramid?
- ▶ Each pound of weight lost on average when participating in a sponsored diet retains 1000 members nationally who would otherwise not renew their membership.
- ▶ Each retained member adds \$100 on profits, so an additional pound taken off earns the chain- $1000 \times 100 = \$100,000$.
- ▶ If Atkins weight reduces weight by 2 pounds more than conventional diet, then resulting increase in membership profits = \$200,000- this is the licensing fee- break even point.
- ▶ Hence the Atkins diet needs to generate at least 2 more pounds in weight loss to make it a profitable choice.

State the Hypothesis

- ▶ In order to be profitable for the fitness chain, the Atkins diet has to win by more than 2 pounds, on average.

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 - ▶ $H_1 : \mu_A - \mu_C > 2$

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- ▶ The Hypothesis is
 - ▶ $H_0 : \mu_A - \mu_C \leq 2$
 - ▶ $H_1 : \mu_A - \mu_C > 2$
- ▶ In order to test and estimate the difference between two population means, we draw random samples from each of two populations. Initially, we will consider independent samples, that is, samples that are completely unrelated to one another.

Sampling Distribution of $\bar{X}_1 - \bar{X}_2$

- ▶ $\bar{X}_1 - \bar{X}_2$ is normally distributed if the original populations are normal or approximately normal if the populations are nonnormal and the sample sizes are large ($n_1, n_2 > 30$)

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- ▶ The standard error is

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- ▶ The standard error is $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
- ▶
$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

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- ▶
$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

In practice, the Z statistic is rarely used since the population variances are unknown. We consider two cases $\sigma_1^2 = \sigma_2^2$ and $\sigma_1^2 \neq \sigma_2^2$

Making Population Inferences with Unknown Population Variances. Case 1: $\sigma_1^2 = \sigma_2^2$:

- Calculate the pooled variance estimate

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

and use the t - statistic defined by

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \text{ with } \nu = n_1 + n_2 - 2 \text{ degrees of}$$

freedom.

- The confidence interval estimator for $\mu_1 - \mu_2$ when the population variances are equal is given by:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

with ν degrees of freedom

Making Population Inferences with Unknown Population Variances. Case 1: $\sigma_1^2 \neq \sigma_2^2$:

- ▶ The t - statistic defined by

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \text{ with } \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \text{ degrees of freedom.}$$

- ▶ The confidence interval estimator for $\mu_1 - \mu_2$ when the population variances are unequal is given by:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

with ν degrees of freedom

Confidence Interval for the Difference

- ▶ Rather than indirectly compare two confidence intervals, a single confidence interval for the difference between the means is more precise. If our interest lies in the difference between the means, then we should make a confidence interval for that difference.
- ▶ The $100(1 - \alpha)\%$ two-sample confidence interval for the difference in means is

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2} se(\bar{x}_1 - \bar{x}_2)$$

Two Independent Samples

- Two samples, each having p measurements where we have a random sample of size n_1 from population 1 and a random sample of size n_2 from population 2.

Sample from population 1

$$\overbrace{X_{11}, X_{12}, \dots, X_{1n_1}}$$

Sample Mean

$$\bar{X}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} x_{1j}$$

Sample Covariance Matrix

$$S_1 = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (x_{1j} - \bar{X}_1)(x_{1j} - \bar{X}_1)'$$

Sample from population 2

$$\overbrace{X_{21}, X_{22}, \dots, X_{2n_2}}$$

Sample Mean

$$\bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} x_{2j}$$

Sample Covariance Matrix

$$S_2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (x_{2j} - \bar{X}_2)(x_{2j} - \bar{X}_2)'$$

- Hypothesis: $\mu_1 = \mu_2$

Assumptions

1. The sample $X_{11}, X_{12}, \dots, X_{1n_1}$ is a random sample of size n_1 from a p -variate population with mean vector μ_1 and covariance matrix Σ_1 .
2. The sample $X_{21}, X_{22}, \dots, X_{2n_2}$ is a random sample of size n_2 from a p -variate population with mean vector μ_2 and covariance matrix Σ_2 .
3. The samples are (statistically) independent of each other. These assumptions are required when we want to test
 - ▶ $H_o : \mu_1 = \mu_2$ or equivalently $\mu_1 - \mu_2 = 0$
 - ▶ $H_A : \mu_1 \neq \mu_2$ or equivalently $\mu_1 - \mu_2 \neq 0$
4. If n_1 and/or n_2 are small, then we must make two additional assumptions:
 - 4.1 Both populations are multivariate normal.
 - 4.2 $\Sigma_1 = \Sigma_2$

Case 1: Known Σ_1 and Σ_2

- ▶ To develop the test for independent populations, we'll start with supposing that we know Σ_1 and Σ_2 (i.e., we don't have to estimate them) and assume first 4 assumptions made on previous slide.
- ▶ $(\bar{X}_1 - \bar{X}_2) \sim N_p \left(\mu_1 - \mu_2, \frac{\Sigma_1}{n_1} + \frac{\Sigma_2}{n_2} \right)$
- ▶ Test statistic:

$$(\bar{X}_1 - \bar{X}_2)' \left(\frac{1}{n_1} \Sigma_1 + \frac{1}{n_2} \Sigma_2 \right)^{-1} (\bar{X}_1 - \bar{X}_2) \sim \chi_p^2$$

Case 2: Σ_1 and Σ_2 Unknown

- ▶ For this more realistic case, we must also assume $\Sigma_1 = \Sigma_2 = \Sigma$
- ▶ Estimate Σ by pooling the data from the two samples:

$$\begin{aligned}
 S_{pool} &= \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1 + n_2 - 2} \\
 &= \frac{\sum_{j=1}^{n_1} (x_{1j} - \bar{X}_1)(x_{1j} - \bar{X}_1)' + \sum_{j=1}^{n_2} (x_{2j} - \bar{X}_2)(x_{2j} - \bar{X}_2)'}{n_1 + n_2 - 2}
 \end{aligned}$$

S_{pool} is an estimator of Σ with $\text{df} = n_1 + n_2 - 2$.

Distribution of Linear Combination

- ▶ $(\bar{X}_1 - \bar{X}_2) \sim N_p \left(\mu_1 - \mu_2, \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \Sigma \right)$
- ▶ Test statistic for $H_0 : \mu_1 - \mu_2 = \delta_0$:

$$T^2 = ((\bar{X}_1 - \bar{X}_2) - \delta_0)' \left(\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pool} \right)^{-1} ((\bar{X}_1 - \bar{X}_2) - \delta_0)$$

Distribution of Test Statistic

- $T^2 = ((\bar{X}_1 - \bar{X}_2) - \delta_0)' \left(\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pool} \right)^{-1} ((\bar{X}_1 - \bar{X}_2) - \delta_0)$
has a sampling distribution that is

$$\frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} F_{p, (n_1 + n_2 - p - 1)}$$

- Reject Null if $T^2 =$
 $((\bar{X}_1 - \bar{X}_2) - \delta_0)' \left(\left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pool} \right)^{-1} ((\bar{X}_1 - \bar{X}_2) - \delta_0) > c^2$
 where $c^2 = \frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} F_{p, (n_1 + n_2 - p - 1)}(\alpha)$

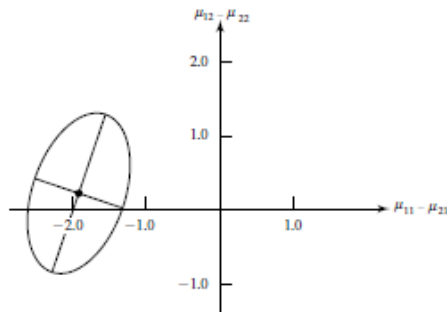
A Simple Example

Fifty bars of soap are manufactured in each of two ways. Two characteristics, X_1 = lather and X_2 = mildness are measured. The summary statistics for bars produced by methods 1 and 2 are

$$\bar{X}_1 = \begin{bmatrix} 8.3 \\ 4.1 \end{bmatrix}, \bar{X}_2 = \begin{bmatrix} 10.2 \\ 3.9 \end{bmatrix}$$
$$S_1 = \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}, S_2 = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

Can we conclude that the two methods of manufacturing soap produce different results.

95% Confidence Ellipse



Clearly, $\mu_1 - \mu_2$ is not in the ellipse, and we conclude that the two methods of manufacturing soap produce different results.

Example: Two Independent Samples (Example 6.4 JW)

Wisconsin home owners without air-conditioning and those with air-conditioning

X_1 = total on-peak consumption of electricity July 1977 (in kilowatts)

X_2 = total off-peak consumption of electricity July 1977 (in kilowatts)

$$\bar{x}_1 = (204.4, 556.6)' \quad \bar{x}_2 = (130.0, 355.0)'$$

$$\text{and} \quad (\bar{x}_1 - \bar{x}_2) = (74.4, 201.6)$$

$$S_1 = \begin{pmatrix} 13825.3 & 23823.4 \\ 23823.4 & 73107.4 \end{pmatrix} \quad S_2 = \begin{pmatrix} 8632.0 & 19616.7 \\ 19616.7 & 55964.5 \end{pmatrix}$$

$$S_{pool} = \frac{44S_1 + 54S_2}{98} = \begin{pmatrix} 10963.7 & 21505.5 \\ 21505.5 & 63661.3 \end{pmatrix}$$

Example Continued

The estimated covariance matrix of $(\bar{x}_1 - \bar{x}_2)$ is

$$\begin{aligned} S_{\bar{x}_1 - \bar{x}_2} &= \left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pool} \\ &= \left(\frac{1}{45} + \frac{1}{55} \right) \begin{pmatrix} 10963.7 & 21505.5 \\ 21505.5 & 63661.3 \end{pmatrix} \\ &= \begin{pmatrix} 442.98 & 868.91 \\ 868.91 & 2572.12 \end{pmatrix} \end{aligned}$$

To test $H_o : \delta = (\mu_1 - \mu_2) = 0$, compute test statistic

$$\begin{aligned} (\bar{x}_1 - \bar{x}_2)' S_{\bar{x}_1 - \bar{x}_2}^{-1} (\bar{x}_1 - \bar{x}_2) &= (74, 201.6) \begin{pmatrix} 442.98 & 868.91 \\ 868.91 & 2572.12 \end{pmatrix}^{-1} \begin{pmatrix} 74.4 \\ 201.6 \end{pmatrix} \\ &= 16.06 \end{aligned}$$

For $\alpha = .05$: $(98(2)/97) \mathcal{F}_{2,97}(.05) = 2.02(3.1) = 6.24$.

100(1 - α)% Confidence Regions

Is the set of all $\delta = \mu_1 - \mu_2$'s such that

$$\frac{n_1 n_2}{n_1 + n_2} ((\bar{x}_1 - \bar{x}_2) - \delta)' S_{pool}^{-1} ((\bar{x}_1 - \bar{x}_2) - \delta) \leq c^2$$

where

$$c^2 = \frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} \mathcal{F}_{p, (n_1 + n_2 - p - 1)}(\alpha)$$

To study the ellipsoid, we can focus on the eigenvalues and eigenvectors of S_{pool} .

The axes of the ellipsoid are

$$(\bar{x}_1 - \bar{x}_2) \pm \sqrt{\lambda_i} \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) c^2} e_i \quad i = 1, \dots, p$$

where λ_i and e_i are the eigenvalues and eigenvectors of S_{pool} .

Example: Confidence Region

The 95% Confidence Region (Ellipse):

The set of all possible $(\mu_1 - \mu_2)$ that satisfy the following equation:

$$((74.4 - \delta_1), (201.6 - \delta_2)) \begin{pmatrix} 442.98 & 868.91 \\ 868.91 & 2572.12 \end{pmatrix}^{-1} \begin{pmatrix} (74.4 - \delta_1) \\ (201.6 - \delta_2) \end{pmatrix} \leq c^2$$

where $c^2 = (98(2)/97)\mathcal{F}_{2,97}(.05) = 2.02(3.1) = 6.26$.

Eigenvalues and Eigenvectors of S_{pool} are

$$\lambda_1 = 71323.426, \quad e_1 = \begin{pmatrix} 0.3356 \\ 0.9420 \end{pmatrix}$$

and

$$\lambda_2 = 3301.572, \quad e_2 = \begin{pmatrix} 0.9420 \\ -0.3356 \end{pmatrix}$$

Computing Axes of the Ellipse

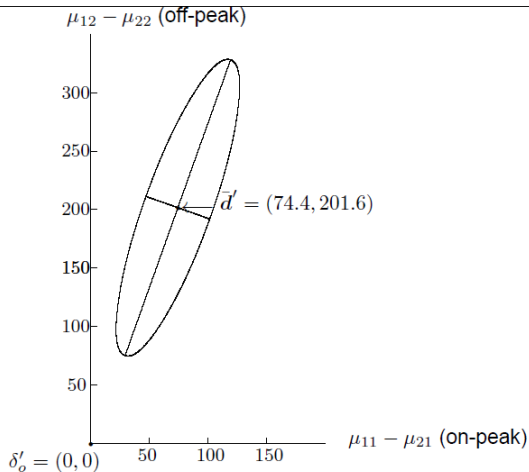
Major axis

$$\begin{aligned}
 \begin{pmatrix} 74.4 \\ 201.6 \end{pmatrix} &\pm \sqrt{\lambda_1} \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) c^2} e_1 \\
 &\pm \sqrt{71323.426} \sqrt{\left(\frac{1}{45} + \frac{1}{55}\right) 6.2441} \begin{pmatrix} 0.3356 \\ 0.9420 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 29.38 & 119.42 \\ 75.24 & 327.96 \end{pmatrix}
 \end{aligned}$$

Minor axis

$$\begin{aligned}
 \begin{pmatrix} 74.4 \\ 201.6 \end{pmatrix} &\pm \sqrt{3301.572} \sqrt{\left(\frac{1}{45} + \frac{1}{55}\right) 6.2441} \begin{pmatrix} 0.9420 \\ -0.3356 \end{pmatrix} \\
 &\rightarrow \begin{pmatrix} 47.21 & 101.59 \\ 211.29 & 191.91 \end{pmatrix}
 \end{aligned}$$

Confidence Region



Simultaneous T^2 Intervals

Let

$$c^2 = \frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} \mathcal{F}_{p, (n_1 + n_2 - p - 1)}(\alpha)$$

With “confidence $100(1 - \alpha)\%$ ”

$$a'(\bar{x}_1 - \bar{x}_2) \pm c \sqrt{a' \left(\frac{1}{n_1} + \frac{1}{n_2} \right) S_{pool} a}$$

will cover $a'(\mu_1 - \mu_2)$ for all possible a .

By appropriate choices for a , we can get component intervals:

$$a_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad a_p = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

Simultaneous T^2 Intervals

So the component intervals are

$$(\bar{x}_{11} - \bar{x}_{21}) \pm c \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) s_{pool,11}}$$

$$(\bar{x}_{12} - \bar{x}_{22}) \pm c \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) s_{pool,22}}$$

$$\vdots \quad \vdots \quad \vdots$$

$$(\bar{x}_{1p} - \bar{x}_{2p}) \pm c \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right) s_{pool,pp}}$$

where

$$c = \sqrt{\frac{(n_1 + n_2 - 2)p}{(n_1 + n_2 - p - 1)} \mathcal{F}_{p, (n_1 + n_2 - p - 1)}(\alpha)}$$

Example: Simultaneous T^2 Intervals

Consider the linear combination vectors:

$$a_1 = (1, 0)' \quad \text{So} \quad a_1' \delta = a_1' (\mu_1 - \mu_2) = \mu_{11} - \mu_{21} = \delta_1$$

and

$$a_2 = (0, 1)' \quad \text{So} \quad a_2' \delta = a_2' (\mu_1 - \mu_2) = \mu_{12} - \mu_{22} = \delta_2$$

Using these we get the intervals for on-peak

$$74.4 \pm (2.502)\sqrt{442.98} \longrightarrow 21.81 \leq \delta_1 \leq 126.99$$

and for off-peak

$$201.6 \pm (2.502)\sqrt{2572.12} \longrightarrow 74.87 \leq \delta_2 \leq 328.33$$

Note: $\sqrt{c^2} = \sqrt{6.26} = 2.502$

Bonferroni and One-at-a-time Intervals

For Bonferroni and One-at-a-Time (i.e., univariate method) intervals, you simply need to change the value of c .

Bonferroni

$$c = t_{n_1+n_2-2}(\alpha/2m)$$

where m = number of intervals formed (probably p , but no more). These should be planned *a priori*.

One-at-a-Time

$$c = t_{n_1+n_2-2}(\alpha/2)$$

Case 3: Large $n_1 - p$ and $n_2 - p$

- ▶ If $n_1 - p$ and $n_2 - p$ are large, then we do not need to assume
 - ▶ $\Sigma_1 = \Sigma_2$
 - ▶ Multivariate normality of x_{1j} and x_{2j}
- ▶ What we need to assume is
 - ▶ Observations between populations are independent.
 - ▶ $(X_{11}, \dots, X_{1,n_1})$ are a random sample from population 1 with μ_1 and Σ_1
 - ▶ $(X_{21}, \dots, X_{2,n_2})$ are a random sample from population 1 with μ_2 and Σ_2
- ▶ For large $n_1 - p$ and $n_2 - p$, the approximate sampling distribution for the test statistic T^2 is χ_p^2

Large Sample Case

- ▶ Estimate the covariance matrix of differences

$$\Sigma_{\bar{X}_1 - \bar{X}_2} = \frac{\Sigma_1}{n_1} + \frac{\Sigma_2}{n_2}$$

using

$$\frac{1}{n_1} S_1 + \frac{1}{n_2} S_2$$

- ▶ Test statistic for $H_0 : \mu_1 - \mu_2 = \delta_0$:

$$T^2 = ((\bar{X}_1 - \bar{X}_2) - \delta_0)' \left(\frac{1}{n_1} S_1 + \frac{1}{n_2} S_2 \right)^{-1} ((\bar{X}_1 - \bar{X}_2) - \delta_0)$$

Case 4: Small Sample with $\Sigma_1 \neq \Sigma_2$

- ▶ Consider whether $\Sigma_1 = \Sigma_2$ is a reasonable assumption.
- ▶ If $n_1 - p$ and $n_2 - p$ are small and $\Sigma_1 \neq \Sigma_2$, then there is no “nice” measure like T^2 whose distribution does not depend on Σ_1 and Σ_2 .
- ▶ We could use Bartlett’s test to test equality of variances, but this assumes data are multivariate normal. Conclusions could be seriously misleading when the populations are nonnormal.
- ▶ Test which is less sensitive to assumption of multivariate normality- Tiku and Balakrishnan.

- └ Paired Comparisons
 - └ Paired Comparisons-Univariate

Outline

Two Independent Samples: Univariate Case

Difference between Two Means

Two Independent Samples: Multivariate

Paired Comparisons

Paired Comparisons-Univariate

Paired Comparisons-Multivariate

Matched Pairs Experiment

- ▶ Previously when comparing two populations, we examined independent samples.
- ▶ If, however, an observation in one sample is matched with an observation in a second sample, this is called a matched pairs experiment.
- ▶ To help understand this concept, let's consider an Example

Example: MBA Job Offers

- ▶ In the last few years a number of web-based companies that offer job placement services have been created.
- ▶ The manager of one such company wanted to investigate the job offers recent MBA's were obtaining.
- ▶ In particular, she wanted to know whether finance majors were being offered higher salaries than marketing majors.
- ▶ In a preliminary study she randomly sampled 50 recently graduated MBAs half of whom majored in finance and half in marketing.
- ▶ From each she obtained the highest salary (including benefits) offer (MBA.xls).

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- ▶ In particular, she wanted to know whether finance majors were being offered higher salaries than marketing majors.
- ▶ In a preliminary study she randomly sampled 50 recently graduated MBAs half of whom majored in finance and half in marketing.
- ▶ From each she obtained the highest salary (including benefits) offer (MBA.xls).
- ▶ Can we infer that finance majors obtain higher salary offers than do marketing majors among MBAs?

Identify

- ▶ The parameter is the difference between two means (where μ_1 = mean highest salary offer to finance majors and μ_2 = mean highest salary offer to marketing majors).
- ▶ Hypothesis:
Because we want to determine whether finance majors are offered higher salaries, the alternative hypothesis will specify that is greater than.
 - ▶ $H_0 : \mu_1 - \mu_2 = 0$
 - ▶ $H_1 : \mu_1 - \mu_2 > 0$
- ▶ Calculation of the F -test of two variances indicates the use the equal-variances test statistic.

- └ Paired Comparisons
 - └ Paired Comparisons-Univariate

Output

t-Test: Two-Sample Assuming Equal Variances		
	<i>Finance</i>	<i>Marketing</i>
Mean	65,624	60,423
Variance	3604,33,294	2622,28,559
Observations	25	25
Pooled Variance	3113,30,926	
Hypothesized Mean Difference	0	
df	48	
t Stat	1.04	
P(T<=t) one-tail	0.1513	
t Critical one-tail	1.6772	
P(T<=t) two-tail	0.3026	
t Critical two-tail	2.0106	

Interpretation

- ▶ The value of the test statistic ($t = 1.04$) and its p -value (.1513) indicate that there is very little evidence to support the hypothesis that finance majors attract higher salary offers than marketing majors.
- ▶ We have some evidence to support the alternative hypothesis, but not enough.

Another way to Experiment

Suppose now that we redo the experiment in the following way.

- ▶ We examine the transcripts of finance and marketing MBA majors.
- ▶ We randomly sample a finance and a marketing major whose grade point average (GPA) falls between 3.92 and 4 (based on a maximum of 4).
- ▶ We then randomly sample a finance and a marketing major whose GPA is between 3.84 and 3.92.
- ▶ We continue this process until the 25th pair of finance and marketing majors are selected whose GPA fell between 2.0 and 2.08.

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- ▶ We then randomly sample a finance and a marketing major whose GPA is between 3.84 and 3.92.
- ▶ We continue this process until the 25th pair of finance and marketing majors are selected whose GPA fell between 2.0 and 2.08.
- ▶ Can we conclude from these data that finance majors draw larger salary offers than do marketing majors?

Observations

- ▶ The experiment described in previous example is one in which the samples are independent. That is, there is no relationship between the observations in one sample and the observations in the second sample.
- ▶ However, in this example the experiment was designed in such a way that each observation in one sample is matched with an observation in the other sample.
- ▶ The matching is conducted by selecting finance and marketing majors with similar GPAs. Thus, it is logical to compare the salary offers for finance and marketing majors in each group.
- ▶ This type of experiment is called **matched pairs** or **paired comparison**

Identify

- For each GPA group, we calculate the matched pair difference between the salary offers for finance and marketing majors.

	A	B	C	D
1	Group	Finance	Marketing	Difference
2	1	95171	89329	5842
3	2	88009	92705	-4696
4	3	98089	99205	-1116
5	4	106322	99003	7319
6	5	74566	74825	-259
25	24	35274	31989	3285
26	25	45835	38788	7047
27	average:	65438.2	60373.68	5064.52
28				
29	Note:	$(65438.20 - 60373.68) = 5064.52$		

- The difference of the means is equal to the mean of the differences, hence we will consider the “mean of the paired differences” as our parameter of interest: $\mu_D = \mu_1 - \mu_2$

Hypothesis and Test Statistic

- ▶ Do Finance majors have higher salary offers than Marketing majors?
 - ▶ $H_0 : \mu_D = 0$
 - ▶ $H_1 : \mu_D > 0$
- ▶ The test statistic for the mean of the population of differences (μ_D) is:

$$t = \frac{\bar{x}_D - \mu_D}{s_D / n_D}$$

Output

	A	B	C
1	t-Test: Paired Two Sample for Means		
2			
3		<i>Finance</i>	<i>Marketing</i>
4	Mean	65,438	60,374
5	Variance	444,981,810	469,441,785
6	Observations	25	25
7	Pearson Correlation	0.9520	
8	Hypothesized Mean Difference	0	
9	df	24	
10	t Stat	3.81	
11	P(T<=t) one-tail	0.0004	
12	t Critical one-tail	1.7109	
13	P(T<=t) two-tail	0.0009	
14	t Critical two-tail	2.0639	

Interpretation

- ▶ The p -value is .0004. There is overwhelming evidence that Finance majors do obtain higher starting salary offers than their peers in Marketing.

Independent Samples or Matched Pairs: Which Experimental Design is Better?

- ▶ The experimental design is an important factor in statistical inference.
- ▶ However, these two examples raise several questions about experimental designs.
 1. Why does the matched pairs experiment result in rejecting the null hypothesis, whereas the independent samples experiment could not?
 2. Should we always use the matched pairs experiment? In particular, are there disadvantages to its use?
 3. How do we recognize when a matched pairs experiment has been performed?

Answers

- ▶ The matched pairs experiment worked in Example by reducing the variation in the data. The numerators of the two test statistics were quite similar. However, the test statistic in Example (Matched pairs) was much larger than the test statistic in Example (Independent samples) because of the standard errors.
- ▶ In the case of comparing two populations of interval data, you must decide whether the samples are independent (in which case the parameter is) or matched pairs (in which case the parameter is) to select the correct test statistic.

Rule of Thumb

- ▶ Does some natural relationship exist between each pair of observations that provides a logical reason to compare the first observation of sample 1 with the first observation of sample 2, the second observation of sample 1 with the second observation of sample 2, and so on?
- ▶ If so, the experiment was conducted by matched pairs. If not, it was conducted using independent samples.

- └ Paired Comparisons
 - └ Paired Comparisons-Multivariate

Outline

Two Independent Samples: Univariate Case

Difference between Two Means

Two Independent Samples: Multivariate

Paired Comparisons

Paired Comparisons-Univariate

Paired Comparisons-Multivariate

Introduction

- ▶ Paired observations arise in a number of different ways:
 - ▶ Every subject (case) responds twice (e.g., pre/post test)
 - ▶ Cases may be matched (on relevant variables) and then randomly assigned to one of two treatments.
 - ▶ Naturally occurring pairs: husbands/wife, siblings, etc.

Notation

- ▶ For $j = 1, \dots, n$ (number of pairs), let
 X_{j1} = measurement of j^{th} case given Treatment 1.
 X_{j2} = measurement of j^{th} case given Treatment 2.
- ▶ We want to examine the differences

$$D_j = X_{j1} - X_{j2}$$

Paired Comparisons-Univariate Case

- By design, the n differences

$$D_j = X_{j1} - X_{j2}$$

- Sampling distribution. If $D_j \sim N(\delta, \sigma_d^2)$, the variable

$$t = \frac{\bar{D} - \delta}{s_d / \sqrt{n}} \sim \text{Student's } t_{n-1} - \text{distribution}$$

$$\text{where } \bar{D} = \frac{1}{n} \sum_{j=1}^n D_j \text{ and } s_d^2 = \frac{1}{n-1} \sum_{j=1}^n (D_j - \bar{D})^2$$

Paired Comparisons-Univariate Case

- ▶ Hypothesis Testing

$$H_0 : \delta = 0$$

$$H_1 : \delta \neq 0$$

- ▶ A $100(1 - \alpha)\%$ confidence interval estimate of δ is given by

$$\bar{D} \pm t_{n-1}(\alpha/2) \frac{s_D}{\sqrt{n}}$$

- ▶ Label p variables for each treatment (condition) within the j^{th} unit for each member of each pair.

$X_{1j1} = \text{Variable 1, treatment 1}$	$X_{2j1} = \text{Variable 1, treatment 2.}$
$X_{1j2} = \text{Variable 2, treatment 1}$	$X_{2j2} = \text{Variable 2, treatment 2.}$
$\vdots$	$\vdots$
$X_{1jp} = \text{Variable } p, \text{ treatment 1}$	$X_{2jp} = \text{Variable } p, \text{ treatment 2.}$
- ▶ the p paired-difference random variables are

$$D_{jp} = X_{1jp} - X_{2jp}$$

Needed for Statistical Inference

Needed for Statistical Inference

- ▶ Assume $D_j \sim N_p(\delta, \Sigma_D)$ and i.i.d for $j = 1, 2, \dots, n$ where

$$\delta = \begin{bmatrix} \delta_1 \\ \delta_2 \\ \vdots \\ \delta_p \end{bmatrix} = E(D_j)$$

- ▶ If the differences $D_1, D_2, \dots, D_n$ are a random sample from $N_p(\delta, \Sigma_D)$ population then

$$T^2 = n(\bar{D} - \delta)' S^{-1} (\bar{D} - \delta) \sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

Statistical Inference

- ▶ Suppose that we have observations $d'_j = (d_{j1}, d_{j2}, \dots, d_{jp})$ for $j = 1, 2, \dots, n$
- ▶ $\bar{d} = \frac{1}{n} \sum_{j=1}^n d_j$
- ▶ $S_d = \frac{1}{n-1} \sum_{j=1}^n (d_j - \bar{d})(d_j - \bar{d})'$
- ▶ Hypothesis Test: $H_0 : \delta = 0$ versus $H_1 : \delta \neq 0$
- ▶ Assumption $D_j \sim N_p(\delta, \Sigma_D)$ are i.i.d
- ▶ Reject H_0 if $T^2 = n\bar{d}S_D^{-1}\bar{d} \geq \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)$

Rejecting H_0

- Confidence Region:

$$n(\bar{D} - \delta)' S^{-1} (\bar{D} - \delta) \leq \frac{(n-1)p}{n-p} \mathcal{F}_{p, n-p}(\alpha)$$

- Simultaneous T^2 Intervals for individual differences of components means

$$\delta_i : \quad \bar{d}_i \pm \sqrt{\frac{(n-1)p}{n-p} \mathcal{F}_{p, n-p}(\alpha)} \sqrt{s_{d_i}^2/n}$$

where $\bar{d}_i$ is mean difference of the i^{th} variable and $s_{d_i}^2$ is the i^{th} diagonal element of S_d .

- Bonferroni $100(1 - \alpha)\%$ confidence intervals

$$\delta_i : \quad \bar{d}_i \pm t_{n-1}(\alpha/2m) \sqrt{s_{d_i}^2/n}$$

where m = the number of confidence intervals (comparisons).

Example

Municipal wastewater treatment plants are required by law to monitor their discharges into rivers and streams on a regular basis. Concern about the reliability of data from one of these self-monitoring programs led to a study in which samples of effluent were divided and sent to two laboratories for testing. One-half of each sample was sent to the Wisconsin State Laboratory of Hygiene, and one-half was sent to a private commercial laboratory routinely used in the monitoring program. Measurements of biochemical oxygen demand (BOD) and suspended solids (SS) were obtained, for $n = 11$ sample splits, from the two laboratories.

Example

Table 6.1 Effluent Data

Sample j	Commercial lab		State lab of hygiene	
	x_{1j1} (BOD)	x_{1j2} (SS)	x_{2j1} (BOD)	x_{2j2} (SS)
1	6	27	25	15
2	6	23	28	13
3	18	64	36	22
4	8	44	35	29
5	11	30	15	31
6	34	75	44	64
7	28	26	42	30
8	71	124	54	64
9	43	54	34	56
10	33	30	29	20
11	20	14	39	21

Source: Data courtesy of S. Weber.

Questions Asked

- ▶ Do the two laboratories' chemical analyses agree?
- ▶ If differences exist, then what is the nature?

Example

$d_{j1} = x_{1j1} - x_{2j1}$	-19	-22	-18	-27	-4	-10	-14	17	9	4	-19
$d_{j2} = x_{1j2} - x_{2j2}$	12	10	42	15	-1	11	-4	60	-2	10	-7

Here

$$\bar{\mathbf{d}} = \begin{bmatrix} \bar{d}_1 \\ \bar{d}_2 \end{bmatrix} = \begin{bmatrix} -9.36 \\ 13.27 \end{bmatrix}, \quad \mathbf{S}_d = \begin{bmatrix} 199.26 & 88.38 \\ 88.38 & 418.61 \end{bmatrix}$$

and

$$T^2 = 11[-9.36, \quad 13.27] \begin{bmatrix} .0055 & -.0012 \\ -.0012 & .0026 \end{bmatrix} \begin{bmatrix} -9.36 \\ 13.27 \end{bmatrix} = 13.6$$

Taking $\alpha = 0.05$, we find that

$p(n-1)/(n-p)F_{p,n-p}(0.5) = 9.47$. $T^2 = 13.6 > 9.47$, we reject H_0 and conclude that there is non zero mean difference between the measurements of the two laboratories.

Simultaneous Confidence Intervals

$$\delta_1: \bar{d}_1 \pm \sqrt{\frac{(n-1)p}{(n-p)} F_{p, n-p}(\alpha)} \sqrt{\frac{s_{d_1}^2}{n}} = -9.36 \pm \sqrt{9.47} \sqrt{\frac{199.26}{11}}$$

or $(-22.46, 3.74)$

$$\delta_2: 13.27 \pm \sqrt{9.47} \sqrt{\frac{418.61}{11}} \quad \text{or} \quad (-5.71, 32.25)$$

Another way to compute T^2

for paired comparisons.

So far we've "divided the sample"; that is, $D = X_1 - X_2$.

Now we'll consider a "Full Sample" method that considers every case as a pair and each with p measures on each member of the pair.

Condition	Pair or "Case" Number					
	1	2	...	j	...	n
(a)	p variables	p variables	...	p variables	...	p variables
(b)	p variables	p variables	...	p variables	...	p variables

So we have $2p$ variables measured for each case (pair). In an experimental situation, the conditions are assumed to have been randomly assigned to members of the pairs.

Full Data Method for Paired Comparisons

Full Data Matrix:

$$\begin{aligned}
 \mathbf{X}_{n \times 2p} &= \left(\begin{array}{cccc|cccc} X_{111} & X_{112} & \cdots & X_{11p} & X_{121} & X_{122} & \cdots & X_{12p} \\ X_{211} & X_{212} & \cdots & X_{21p} & X_{221} & X_{222} & \cdots & X_{22p} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ X_{n11} & X_{n12} & \cdots & X_{n1p} & X_{n21} & X_{n22} & \cdots & X_{n2p} \end{array} \right) \\
 &= \underbrace{(\mathbf{X}_1)}_{n \times p} \mid \underbrace{(\mathbf{X}_2)}_{n \times p}
 \end{aligned}$$

Full Sample Mean Vector:

$$\bar{\mathbf{X}}' = (\bar{X}_{11}, \bar{X}_{12}, \dots, \bar{X}_{1p} \mid \bar{X}_{21}, \dots, \bar{X}_{2p}) = (\bar{\mathbf{X}}_1' \mid \bar{\mathbf{X}}_2')$$

Full Data Method for Paired Comparisons

Full Data Sample Covariance Matrix:

$$S_{2p \times 2p} = \left(\begin{array}{c|c} S_{11} & S_{12} \\ \hline S_{21} & S_{22} \end{array} \right)$$

where

S_{11} is the $(p \times p)$ covariance matrix for X_1

S_{22} is the $(p \times p)$ covariance matrix for X_2

$S_{12} = S'_{21}$ is the $(p \times p)$ covariance matrix between X_1 & X_2 .

Define a **Contrast Matrix**:

$$\begin{aligned} C_{p \times 2p} &= \left(\begin{array}{cccc|cccc} 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & 0 & \cdots & -1 \end{array} \right) \\ &= (I_{p \times p} | -I_{p \times p}) \end{aligned}$$

Computations for Full Data

Let $x_{j,(2p \times 1)}$ = j row of $X_{(n \times 2p)}$ written as a column vector

$$d_j = Cx_j$$

$$\bar{d} = C\bar{x} = C \frac{1}{n} \sum_{j=1}^n x_j$$

Putting all of this together yields

$$T^2 = n\bar{x}' C' (CSC')^{-1} C\bar{x}$$

MS 6015: Multivariate Statistical Methods for Business

ANOVA

Reference: Keller: Chapter 14

Outline

Motivation

One-Way Analysis of Variance

Two Way Analysis of Variance

Two-Factor Analysis of Variance

Learning Objectives

- ▶ Analysis of variance is a technique that allows us to compare two or more populations of interval data.
- ▶ Analysis of variance is:
 - ▶ an extremely powerful and widely used procedure.
 - ▶ a procedure which determines whether differences exist between population means.
 - ▶ a procedure which works by analyzing **sample variance**.

Motivation

- ▶ Do higher average returns imply better performance?
- ▶ Why do two strategies with the same mean lead to different conclusions?
- ▶ ANOVA answers: is the difference in means large relative to variability?

Case 1: Stable Funds (Low Variance)

Monthly returns (%):

Fund A	Fund B	Fund C
5.90	5.51	5.01
5.92	5.50	5.00
5.91	5.50	4.99
5.89	5.49	4.98
5.88	5.50	5.02

Means: 5.9, 5.5, 5.0

Variances: extremely small

Interpretation (Case 1)

- ▶ Returns are very consistent
- ▶ Market noise is minimal
- ▶ Differences in means stand out

ANOVA Result

Very large $F \Rightarrow$ statistically significant

Finance meaning: evidence of true manager skill.

Case 2: Volatile Funds (High Variance)

Monthly returns (%):

Fund A	Fund B	Fund C
5.90	6.31	4.52
4.42	3.54	6.93
7.51	4.73	4.48
7.89	7.20	5.55
3.78	5.72	3.52

Means: 5.9, 5.5, 5.0

Variances: very large

Interpretation (Case 2)

- ▶ Returns fluctuate heavily
- ▶ High market noise
- ▶ Mean differences unreliable

ANOVA Result

Small $F \Rightarrow$ not significant

Finance meaning: performance could be luck.

Key Idea

ANOVA compares:

$$F = \frac{\text{Between-group variation (signal)}}{\text{Within-group variation (noise)}}$$

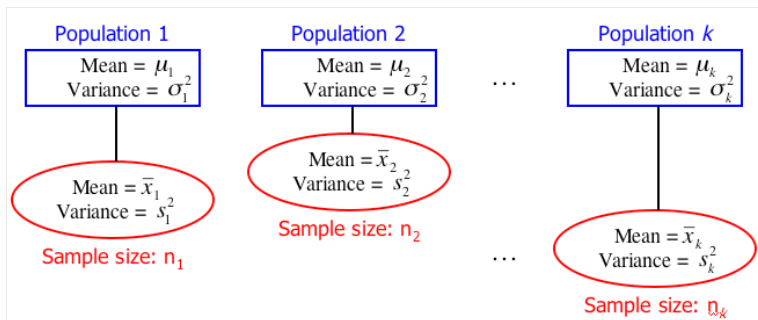
- ▶ Large $F \Rightarrow$ real differences
- ▶ Small $F \Rightarrow$ differences likely due to randomness

Finance Mapping

Statistics	Finance
Between-group variation	Manager skill
Within-group variation	Market volatility (risk)
F-statistic	Performance significance
Variance	Risk

One-Way Analysis of Variance

- Independent samples are drawn from k populations:



These populations are referred to as treatments

It is not a requirement that $n_1 = n_2 = \dots = n_k$

Terminology

- ▶ x is the response variable, and its values are responses.
- ▶ x_{ij} refers to the i^{th} observation in the j^{th} sample. E.g. x_{35} is the third observation of the fifth sample.
- ▶ The grand mean, $\bar{\bar{x}}$, is the mean of all the observations, i.e.:

$$\bar{\bar{x}} = \frac{\sum_{j=1}^k \sum_{i=1}^{n_j} x_{ij}}{n}; n = n_1 + n_2 + \dots + n_j$$

- ▶ Population classification criterion is called a **factor**.
- ▶ Each population is a **factor level**.

Example

- ▶ In the last decade stockbrokers have drastically changed the way they do business. It is now easier and cheaper to invest in the stock market than ever before.
- ▶ What are the effects of these changes?
- ▶ To help answer this question a financial analyst randomly sampled 366 American households and asked each to report the age of the head of the household and the proportion of their financial assets that are invested in the stock market.

Categories

- ▶ The age categories are
 - ▶ Young (Under 35)
 - ▶ Early middle-age (35 to 49)
 - ▶ Late middle-age (50 to 65)
 - ▶ Senior (Over 65)
- ▶ The analyst was particularly interested in determining whether the ownership of stocks varied by age.

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Do these data allow the analyst to determine that there are differences in stock ownership between the four age groups?

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- ▶ The age category is the factor we're interested in. This is the only factor under consideration (hence the term “one way” analysis of variance).
- ▶ Each population is a factor level. In this example, there are four factor levels: Young, Early middle age, Late middle age, and Senior.

Identify

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 H_1 : at least two means differ
- ▶ OK. Now we need some test statistics...

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- ▶ A large SST indicates large variation between sample means which supports H_1 .

Compute

- ▶ The following sample statistics and grand mean were computed
 - ▶ $\bar{x}_1 = 44.40$
 - ▶ $\bar{x}_2 = 52.47$
 - ▶ $\bar{x}_3 = 51.14$
 - ▶ $\bar{x}_4 = 51.84$
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- ▶ Is $SST = 3,741.4$ "large enough"?

Equal Variances Two Sample Test Statistic

- ▶ When we performed the equal-variances test to determine whether two means differed we used

$$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

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- ▶ The numerator measures the difference between sample means and the denominator measures the variation in the samples.

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- ▶ In the second formulation, it is easier to see that it provides a **measure of the amount of variation** we can expect from the random variable we've observed.

Going Back to Example: Compute

► Since:
$$SST = \sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2$$

If it were the case that: $\bar{x}_1 = \bar{x}_2 = \bar{x}_3 = \bar{x}_4$ then $SST = 0$ and our null hypothesis, $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ would be supported.

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- More generally, a small value of SST supports the null hypothesis. A large value of SST supports the alternative hypothesis.
- The question is, how large is “large enough”?

Compute

- ▶ Hence, the between-treatments variation, sum of squares for treatments, is

$$SST = 84(44.40 - 50.18)^2 + \dots + 58(51.84 - 50.18)^2 = 3714.4$$

- ▶ We calculate the sample variances as:

$s_1^2 = 386.55$, $s_2^2 = 469.44$, $s_3^2 = 471.82$, $s_4^2 = 444.79$ and from these, calculate the within-treatments variation (sum of squares for error) as:

$$SSE = (84 - 1)386.55 + \dots + (58 - 1)444.79 = 161,871.0$$

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- ▶ We still need a couple more quantities in order to relate SST and SSE together in a meaningful way?

Mean Squares

- ▶ The mean square for treatments (MST) is given by:

$$MST = \frac{SST}{k-1}$$

- ▶ The mean square for errors (MSE) is given by:

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- ▶ The Test statistic:

$$F = \frac{MST}{MSE}$$

is F -distributed with $k - 1$ and $n - k$ degrees of freedom.

Example: Compute F

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- ▶ The F -statistic is $F = \frac{MST}{MSE} = \frac{1247.12}{447.16} = 2.79$
- ▶ Does $F = 2.79$ fall into a rejection region or not? What is the p -value?

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- ▶ Since the purpose of calculating the F -statistic is to determine whether the value of SST is large enough to reject the null hypothesis, if SST is large, F will be large.

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Interpret

- ▶ Since the p -value is .0405, which is small we reject the null hypothesis ($H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$) in favor of the alternative hypothesis (H_1 : at least two population means differ).
- ▶ That is: there is enough evidence to infer that the mean percentages of assets invested in the stock market differ between the four age categories

ANOVA Table

The results of **analysis of variance** are usually reported in an ANOVA table:

Source of Variation	degrees of freedom	Sum of Squares	Mean Square
Treatments	$k-1$	SST	$MST = SST / (k-1)$
Error	$n-k$	SSE	$MSE = SSE / (n-k)$
Total	$n-1$	SS(Total)	F-stat = MST / MSE

ANOVA and t -tests of 2 means

- ▶ Why do we need the analysis of variance? Why not test every pair of means? For example say $k = 6$.
- ▶ There are $6_{C_2} = 6(5)/2 = 15$ different pairs of means.
 - ▶ 1&2 1&3 1&4 1&5 1&6
 - ▶ 2&3 2&4 2&5 2&6
 - ▶ 3&4 3&5 3&6
 - ▶ 4&5 4&6
 - ▶ 5&6

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 - ▶ 5&6
- ▶ If we test each pair with $\alpha = .05$ we increase the probability of making a Type I error. If there are no differences then the probability of making at least one Type I error is $1 - (.95)^{15} = 1 - .463 = .537$

ANOVA versus t -tests

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ANOVA versus t -tests

- ▶ We have argued that the t -tests cannot replace analysis of variance
- ▶ Can we use the analysis of variance instead of t -tests?
 - ▶ Anova allows us to test for a difference only.
 - ▶ Anova requires equal population variances.

Checking the Required Conditions

- ▶ The F -test of the analysis of variance requires that the random variable be normally distributed with **equal variances**. The normality requirement is easily checked graphically by producing the histograms for each sample.

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- ▶ The F -test of the analysis of variance requires that the random variable be normally distributed with **equal variances**. The normality requirement is easily checked graphically by producing the histograms for each sample.
- ▶ The equality of variances is examined by printing the sample standard deviations or variances. The similarity of sample variances allows us to assume that the population variances are equal.

Violation of the Required Conditions

- ▶ If the data are not normally distributed we can replace the one-way analysis of variance with its nonparametric counterpart, which is the Kruskal-Wallis test.

Violation of the Required Conditions

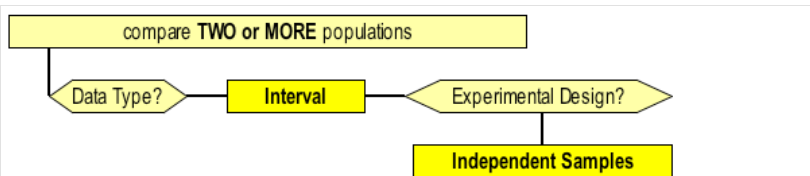
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Violation of the Required Conditions

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- ▶ If the population variances are unequal, we can use several methods to correct the problem.
- ▶ However, these corrective measures are beyond the level of the course.

In Summary: Identifying Factors

Factors that Identify the One-Way Analysis of Variance:



Parameters of interest: $\mu_1, \mu_2, \dots, \mu_k$

Test Statistic:

$$F = \frac{MST}{MSE}$$

Required Conditions: Populations are normal with equal variances.

If populations are non-normal, apply Kruskal-Wallis test.

Multiple Comparisons

- ▶ When we conclude from the one-way analysis of variance that at least two treatment means differ (i.e. we reject the null hypothesis that $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$), we often need to know which treatment means are responsible for these differences.

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- ▶ We will examine three statistical inference procedures that allow us to determine which population means differ: ?
 - ▶ Fisher's least significant difference (LSD) method
 - ▶ Bonferroni adjustment, and
 - ▶ Tukey's multiple comparison method.

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IF $|\bar{x}_i - \bar{x}_j| > N_{Critical}$
THEN we conclude μ_i and μ_j differ.
- ▶ The larger sample mean is then believed to be associated with a larger population mean.

Independent Samples and Blocks

- ▶ A **randomized block design** experiment reduces the variation **within** the samples, making it easier to detect differences between populations.

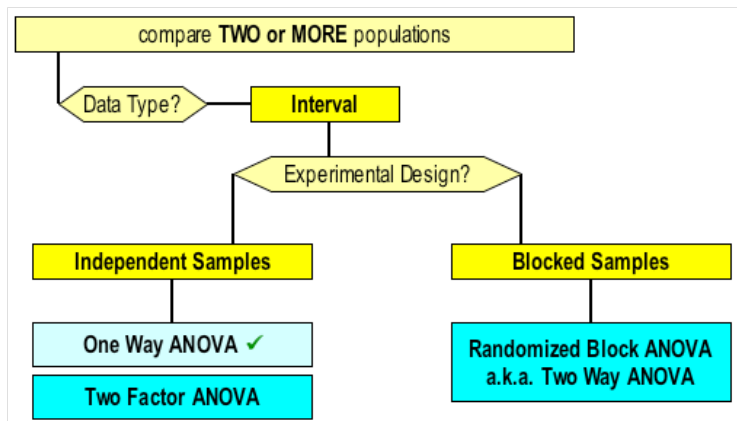
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- ▶ A **randomized block design** experiment reduces the variation **within** the samples, making it easier to detect differences between populations.
- ▶ The term **block** refers to a matched group of observations from each population.
- ▶ We can also perform a blocked experiment by using the same subject for each treatment in a “**repeated measures**” experiment.

- ▶ The randomized block experiment is also called the two-way analysis of variance, not to be confused with the two-factor analysis of variance. To illustrate where we're headed



Randomized Block Analysis of Variance

- ▶ The purpose of designing a randomized block experiment is to reduce the within-treatments variation to more easily detect differences between the treatment means.

Randomized Block Analysis of Variance

- ▶ The purpose of designing a randomized block experiment is to reduce the within-treatments variation to more easily detect differences between the treatment means.
- ▶ In this design, we partition the total variation into three sources of variation:

$$SS(\text{Total}) = SST + SSB + SSE$$

where SSB , the **sum of squares for blocks**, measures the variation between the blocks.

Randomized Blocks

- In addition to k treatments, we introduce notation for b blocks in our experimental design

	Treatments				
Block	1	2	...	k	Block Mean
1	x_{11}	x_{12}	...	x_{1k}	$\bar{x}[B]_1$
2	x_{21}	x_{22}	...	x_{2k}	$\bar{x}[B]_2$
:	:	:		:	:
b	x_{b1}	x_{b2}	...	x_{bk}	$\bar{x}[B]_b$
Treatment Mean	$\bar{x}[T]_1$	$\bar{x}[T]_2$	...	$\bar{x}[T]_k$	

Sum of Squares : Randomized Block

Squaring the 'distance' from the grand mean, leads to the following set of formulae

$$SS(Total) = \sum_{j=1}^k \sum_{i=1}^b (x_{ij} - \bar{\bar{x}})^2$$

$$SST = \sum_{j=1}^k b(\bar{x}[T]_j - \bar{\bar{x}})^2$$

$$SSB = \sum_{i=1}^b k(\bar{x}[B]_i - \bar{\bar{x}})^2$$

$$SSE = \sum_{j=1}^k \sum_{i=1}^b (x_{ij} - \bar{x}[T]_j - \bar{x}[B]_i + \bar{\bar{x}})^2$$

$$MST = \frac{SST}{k-1}$$

$$MSB = \frac{SSB}{b-1}$$

$$MSE = \frac{SSE}{n-k-b+1}$$

$$F = \frac{MST}{MSE}, \quad v_1 = k-1 \quad \& \quad v_2 = n-k-b+1 \quad d.f.$$

$$F = \frac{MSB}{MSE}, \quad v_1 = b-1 \quad \& \quad v_2 = n-k-b+1 \quad d.f.$$

Anova Table

We can summarize this new information in an analysis of variance (ANOVA) table for the randomized block analysis of variance as follows

Source of Variation	d.f.:	Sum of Squares	Mean Square	<i>F</i> Statistic
Treatments	$k-1$	SST	$MST = SST/(k-1)$	$F = MST/MSE$
Blocks	$b-1$	SSB	$MSB = SSB/(b-1)$	$F = MSB/MSE$
Error	$n-k-b+1$	SSE	$MSE = SSE/(n-k-b+1)$	
Total	$n-1$	SS(Total)		

Example

- ▶ Many North Americans suffer from high levels of cholesterol, which can lead to heart attacks. For those with very high levels (over 280), doctors prescribe drugs to reduce cholesterol levels.
- ▶ A pharmaceutical company has recently developed four such drugs. To determine whether any differences exist in their benefits, an experiment was organized.
- ▶ The company selected 25 groups of four men, each of whom had cholesterol levels in excess of 280. In each group, the men were matched according to age and weight.
- ▶ The drugs were administered over a 2-month period, and the reduction in cholesterol was recorded.

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- ▶ The company selected 25 groups of four men, each of whom had cholesterol levels in excess of 280. In each group, the men were matched according to age and weight.
- ▶ The drugs were administered over a 2-month period, and the reduction in cholesterol was recorded.
- ▶ Do these results allow the company to conclude that differences exist between the four new drugs?

Identify

- ▶ Each of the four drugs can be considered a treatment.
- ▶ Each group) can be blocked, because they are matched by age and weight.
- ▶ By setting up the experiment this way, we eliminates the variability in cholesterol reduction related to different combinations of age and weight. This helps **detect differences** in the mean cholesterol reduction **attributed to the different** drugs.

Checking the Required Conditions

- ▶ The F -test of the randomized block design of the analysis of variance has the same requirements as the independent samples design.
- ▶ That is, the random variable must be normally distributed and the population variances must be equal.
- ▶ The histograms appear to support the validity of our results; the reductions appear to be normal.
- ▶ The equality of variances requirement also appears to be met.

Developing an Understanding of Statistical Concepts

- ▶ The randomized block experiment is an extension of the matched pairs experiment.
- ▶ In the matched pairs experiment, we simply remove the effect of the variation caused by differences between the experimental units. The effect of this removal is seen in the decrease in the value of the standard error and the increase in the value of the t -statistic.

Developing an Understanding of Statistical Concepts

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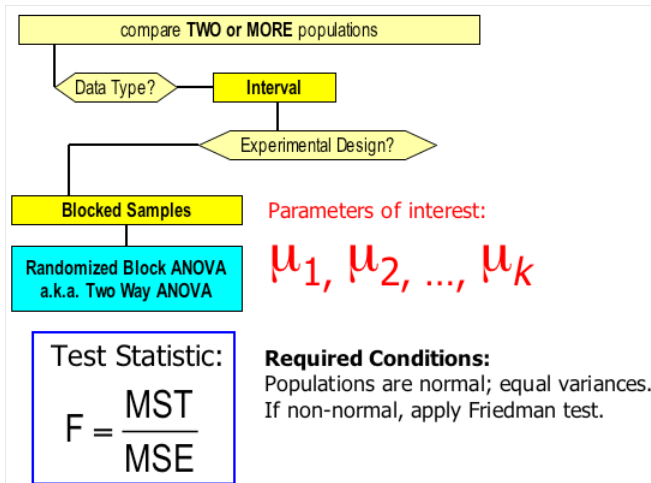
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- ▶ In the randomized block experiment of the analysis of variance, we actually measure the variation between the blocks by computing SSB.
- ▶ The sum of squares for error is reduced by SSB, making it easier to detect differences between the treatments.
- ▶ Additionally, we can test to determine whether the blocks differ—a procedure we were unable to perform in the matched pairs experiment.

Summary:RBD

Factors that Identify the Randomized Block of the Analysis of Variance



Analysis of Variance Experimental Designs

- ▶ A **multifactor experiment** is one where there are two or more factors that define the treatments.

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Analysis of Variance Experimental Designs

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- ▶ For example, if instead of just varying the advertising strategy for our new apple juice product we also varied the advertising medium (e.g. television or newspaper), then we have a two-factor analysis of variance situation.
- ▶ The first factor, advertising strategy, has three levels (convenience, quality, and price) while the second factor, advertising medium, has two levels (TV or print).

Two-Factor Analysis of Variance

- ▶ In factorial experiments, we can examine the effect on the response variable of two or more factors.
- ▶ We can use the analysis of variance to determine whether the Levels of each factor are different from one another.

Example

- ▶ One measure of the health of a nation's economy is how quickly it creates jobs.
- ▶ One aspect of this issue is the number of jobs individuals hold.
- ▶ As part of a study on job tenure, a survey was conducted wherein Americans aged between 37 and 45 were asked how many jobs they have held in their lifetimes.
- ▶ Also recorded were gender and educational attainment.

Example- Continued

- ▶ The categories are
 - ▶ Less than high school (E1)
 - ▶ High school (E2)
 - ▶ Some college/university but no degree (E3)
 - ▶ At least one university degree (E4)
- ▶ The data were recorded for each of the eight categories of Gender and education.

Example- Continued

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 - ▶ High school (E2)
 - ▶ Some college/university but no degree (E3)
 - ▶ At least one university degree (E4)
- ▶ The data were recorded for each of the eight categories of Gender and education.
- ▶ Can we infer that differences exist between genders and educational levels?

Analysis

- ▶ We begin by treating this example as a one-way analysis of Variance with eight treatments.
- ▶ However, the treatments are defined by two different factors.
- ▶ One factor is gender, which has two levels.
- ▶ The second factor is educational attainment, which has four levels.

One Way Anova Results

	A	B	C	D	E	F	G
1	Anova: Single Factor						
2							
3	SUMMARY						
4	<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>		
5	Male E1	10	126	12.60	8.27		
6	Male E2	10	110	11.00	8.67		
7	Male E3	10	106	10.60	11.60		
8	Male E4	10	90	9.00	5.33		
9	Female E1	10	115	11.50	8.28		
10	Female E2	10	112	11.20	9.73		
11	Female E3	10	94	9.40	16.49		
12	Female E4	10	81	8.10	12.32		
13							
14							
15	ANOVA						
16	<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
17	Between Groups	153.35	7	21.91	2.17	0.0467	2.1397
18	Within Groups	726.20	72	10.09			
19							
20	Total	879.55	79				

- ▶ The value of the test statistic is $F = 2.17$ with a p -value of .0467.
- ▶ We conclude that there are differences in the number of jobs between the eight treatments.

Questions?

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- ▶ Namely, can we conclude that the differences in the mean number of jobs are caused by differences between males and females?
- ▶ Or are they caused by differences between educational levels?
- ▶ Or, perhaps, are there combinations, called interactions of gender and education that result in especially high or low numbers?

Factorial Experiment

- ▶ A complete **factorial experiment** is an experiment in which the data for all possible combinations of the levels of the factors are gathered. This is also known as a two-way classification.
- ▶ The two factors are usually labeled A & B , with the number of levels of each factor denoted by a & b respectively.
- ▶ The number of observations for each combination is called a **replicate**, and is denoted by r . For our purposes, the number of replicates will be the same for each treatment, that is they are **balanced**.

Factorial Design

	Male	Female
Less than high school	10 9 12 16 14 17 13 9 11 15	7 13 14 6 11 14 13 11 14 12
High School	12 11 9 14 12 16 10 10 5 11 15	7 12 6 15 10 13 9 15 12 13 5
Less than Bachelor's degree	8 7 7 7 9 14 15 11 13	13 12 3 13 11 15 5 9 8
At least one Bachelor's degree	8 9 5 11 13 8 7 11 10 8	7 9 3 7 9 6 10 15 4 11

Example: Terminology

- ▶ Thus, we use a complete factorial experiment where the number of treatments is ab with r replicates per treatment.
- ▶ In Example, $a = 2$, $b = 4$, and $r = 10$.
- ▶ As a result, we have 10 observations for each of the eight treatments.

Sum of Squares

- ▶ If you examine the ANOVA table, you can see that the total variation is $SS(Total) = 879.55$, the sum of squares for treatments is $SST = 153.35$, and the sum of squares for error is $SSE = 726.20$.
- ▶ The variation caused by the treatments is measured by SST .
- ▶ In order to determine whether the differences are due to factor A , factor B , or some interaction between the two factors, we need to partition SST into three sources.

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- ▶ The variation caused by the treatments is measured by SST .
- ▶ In order to determine whether the differences are due to factor A , factor B , or some interaction between the two factors, we need to partition SST into three sources. These are $SS(A)$, $SS(B)$, and $SS(AB)$.

Anova Table

Source of Variation	d.f.:	Sum of Squares	Mean Square	F Statistic
Factor A	a-1	SS(A)	$MS(A)=SS(A)/(a-1)$	$F=MS(A)/MSE$
Factor B	b-1	SS(B)	$MS(B)=SS(B)/(b-1)$	$F=MS(B)/MSE$
Interaction	$(a-1)(b-1)$	SS(AB)	$MS(AB) = \frac{SS(AB)}{[(a-1)(b-1)]}$	$F=MS(AB)/MSE$
Error	n-ab	SSE	$MSE=SSE/(n-ab)$	
Total	n-1	SS(Total)		

Going Back to the Example

- ▶ Test for the differences between the Levels of Factor A
 - ▶ H_0 : The means of the a levels of Factor A are equal
 - ▶ H_1 : At least two means differ
 - ▶ Test statistic: $F = MS(A)/MSE$

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Testing for Other Factor

- ▶ Test for the differences between the Levels of Factor B
 - ▶ H_0 : The means of the a levels of Factor B are equal
 - ▶ H_1 : At least two means differ
 - ▶ Test statistic: $F = MS(B)/MSE$
- ▶ Example

Testing for Other Factor

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 - ▶ H_1 : At least two means differ
 - ▶ Test statistic: $F = MS(B)/MSE$
- ▶ Example : Are there differences in the mean number of jobs between the four educational levels?
 - ▶ $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$
 - ▶ H_1 : At least two means differ

Test for Interaction

- ▶ Test for interaction between Factors A and B
 - ▶ H_0 : Factors A and B do not interact to affect the mean responses.
 - ▶ H_1 : Factors A and B do interact to affect the mean responses.
 - ▶ Test statistic: $F = MS(AB)/MSE$
- ▶ Example

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 - ▶ H_0 : Factors A and B do not interact to affect the mean responses.
 - ▶ H_1 : Factors A and B do interact to affect the mean responses.
 - ▶ Test statistic: $F = MS(AB)/MSE$
- ▶ Example : Are there differences in the mean sales caused by interaction between gender and educational level?

Interpret Results

- ▶ There are significant differences between the mean number of jobs held by people with different educational backgrounds.

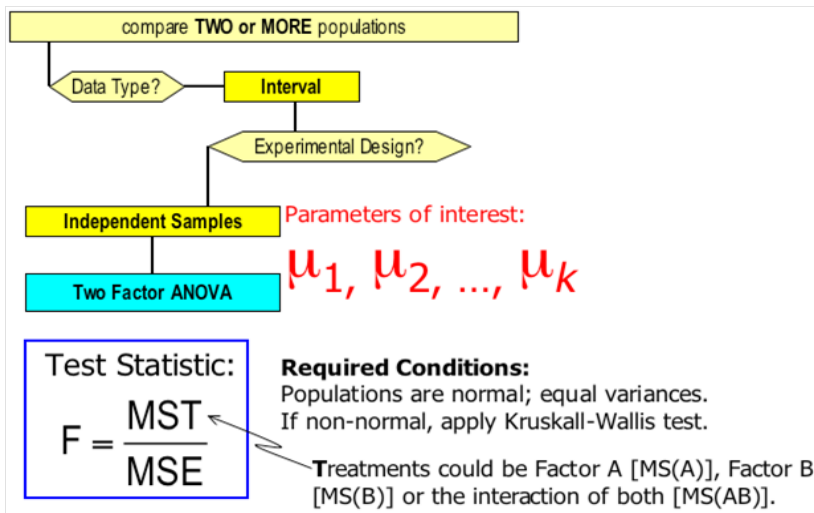
Interpret Results

- ▶ There are significant differences between the mean number of jobs held by people with different educational backgrounds.
- ▶ There is no difference between the mean number of jobs held by men and women.

Interpret Results

- ▶ There are significant differences between the mean number of jobs held by people with different educational backgrounds.
- ▶ There is no difference between the mean number of jobs held by men and women.
- ▶ Finally, there is no interaction.

Two Factor Summary



Anova Summary

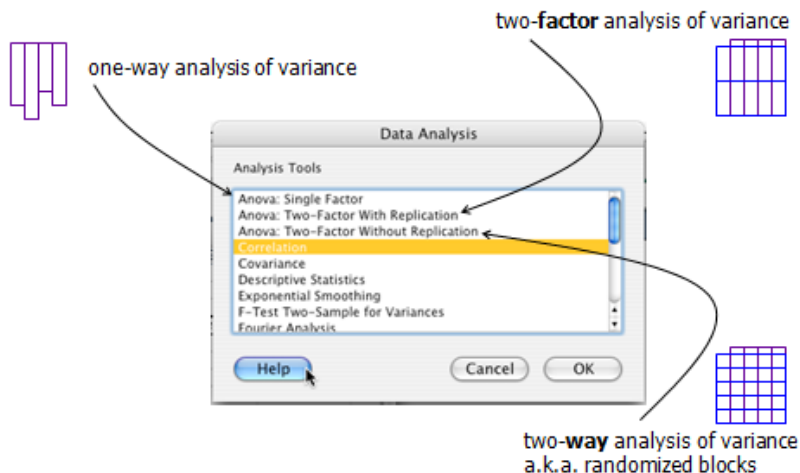


Figure:

MS 6015: Multivariate Statistical Methods for Business

Comparing Several Multivariate Population Means: MANOVA ¹

¹Reference: JW Chapter 6, Section 4-10

From Last Class- Comparing Means

- ▶ Independent Samples Test
- ▶ Paired Comparison

Outline

One-Way Analysis of Variance

One-Way Manova

Two Way Analysis of Variance

Two Way MANOVA

Two-Factor Multivariate Analysis of Variance

Univariate Two-Way Fixed Effects Model

Business Question

A company compares 3 training programs for employees.
They measure:

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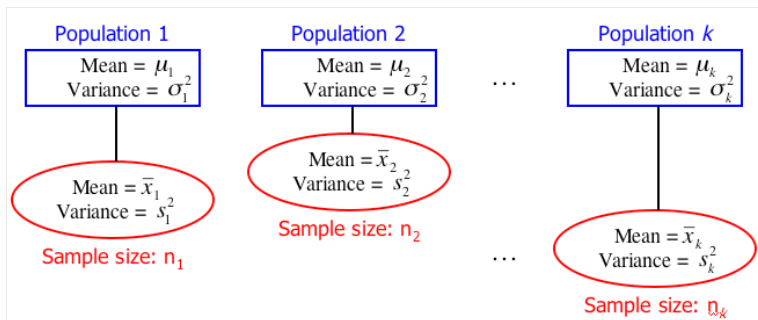
Can we compare programs using only ONE variable?

No.

Because performance is multidimensional.

One-Way Analysis of Variance

- Independent samples are drawn from k populations:



These populations are referred to as treatments

It is not a requirement that $n_1 = n_2 = \dots = n_k$

Data Table

		Treatment			
		1	2	...	k
Subject	1	X_{11}	X_{21}	...	X_{k1}
	2	X_{12}	X_{22}		X_{k2}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	n_i	X_{1n_1}	X_{2n_2}	...	X_{kn_k}

Terminology

- ▶ X is the response variable, and its values are responses.
- ▶ X_{lj} refers to the j^{th} observation in the l^{th} treatment. E.g. X_{35} is the fifth observation of the third treatment.
- ▶ The grand mean, $\bar{X}$, is the mean of all the observations, i.e.:

$$\bar{X} = \frac{\sum_{l=1}^k \sum_{j=1}^{n_l} X_{lj}}{n}; n = n_1 + n_2 + \dots n_k$$

- ▶ Population classification criterion is called a **factor**.
- ▶ Each population is a **factor level**.

One-Way ANOVA

The univariate case where $p = 1$

- ▶ Assumptions: $X_{lj} \sim N(\mu_l, \sigma^2)$ i.i.d FOR $j = 1, \dots, n_l$ and $l = 1, 2, \dots, k$

- ▶ Hypothesis: $H_o : \mu_1 = \mu_2 = \dots = \mu_k$ versus H_1 : not H_o

- ▶ $\underbrace{\mu_l}_{l^{th} \text{ population mean}} = \underbrace{\mu}_{\text{grand mean}} + \underbrace{\mu_l - \mu}_{l^{th} \text{ treatment effect}}$

$$= \mu + \tau_l$$

- ▶ If $\mu_1 = \mu_2 = \dots = \mu_k$, then an equivalent way to write the null hypothesis is

$$H_o : \tau_1 = \tau_2 = \dots = \tau_k = 0$$

The Model for an Observation

$$X_{lj} = \mu + \tau_j + \epsilon_{lj}$$

where $\epsilon_{lj} \sim N(0, \sigma^2)$ and independent.

- ▶ ϵ_{lj} is random error

- ▶ Identification constraint: $\sum_{l=1}^k \tau_l = 0$

- ▶ An analogous decomposition of an observation is

$$\underbrace{x_{lj}}_{\text{observation}} = \underbrace{\bar{x}}_{\substack{\text{overall sample} \\ \text{mean}}} + \underbrace{(\bar{x}_l - \bar{x})}_{\substack{\text{estimated} \\ \text{treatment} \\ \text{effect}}} + \underbrace{(x_{lj} - \bar{x}_l)}_{\text{residual error}}$$

$\bar{x}$ is the estimator of μ ; $\hat{\tau}_l = (\bar{x}_l - \bar{x})$ is estimator of τ_l and $(x_{lj} - \bar{x}_l)$ is estimator of ϵ_{lj}

The Sum of Squares

- Sum of Squared Observations

$$SS_{obs} = SS_{total} = \sum_{l=1}^k \sum_{j=1}^{n_l} X_{lj}^2$$

- The three components of X_{lj} can be taken and sum of squares can be formed as

$$\text{► } SS_{mean} = \sum_{l=1}^k \sum_{j=1}^{n_l} \bar{X}^2 = \left(\sum_{l=1}^k n_l \right) \bar{X}^2$$

$$\text{► } SS_{treatment} = \sum_{l=1}^k \sum_{j=1}^{n_l} \hat{\tau}_l^2 = \sum_{l=1}^k \sum_{j=1}^{n_l} (\bar{X}_l - \bar{X})^2 = \sum_{l=1}^k n_l (\bar{X}_l - \bar{X})^2$$

$$\text{► } SS_{residual} = \sum_{l=1}^k \sum_{j=1}^{n_l} \hat{\epsilon}_{ij}^2 = \sum_{l=1}^k \sum_{j=1}^{n_l} (X_{lj} - \bar{X}_l)^2$$

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- ▶ Since $\mu_1 = \mu_2 = \mu_3 = \mu_4$ is of interest to us, a statistic that measures the **proximity of the sample means to each other** would also be of interest.

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- ▶ A large SST indicates large variation between sample means which supports H_1 .

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- ▶ In the second formulation, it is easier to see that it provides a **measure of the amount of variation** we can expect from the random variable we've observed.

Mean Squares

- ▶ The mean square for treatments (MST) is given by:

$$MST = \frac{SST}{k-1}$$

- ▶ The mean square for errors (MSE) is given by:

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- ▶ The Test statistic:

$$F = \frac{MST}{MSE}$$

is F -distributed with $k - 1$ and $n - k$ degrees of freedom.

- ▶ Since the purpose of calculating the F -statistic is to determine whether the value of SST is large enough to reject the null hypothesis, if SST is large, F will be large.

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- ▶ Since the purpose of calculating the F -statistic is to determine whether the value of SST is large enough to reject the null hypothesis, if SST is large, F will be large.
- ▶ $p - \text{value} = P(F > F_{\text{stat}})$

What ANOVA is Actually Measuring

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Random variation between individuals

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Random variation between individuals

Large $F \Rightarrow$ treatments are different

ANOVA Table

The results of **analysis of variance** are usually reported in an ANOVA table:

Source of Variation	degrees of freedom	Sum of Squares	Mean Square
Treatments	$k-1$	SST	$MST = SST / (k-1)$
Error	$n-k$	SSE	$MSE = SSE / (n-k)$
Total	$n-1$	SS(Total)	F-stat = MST / MSE

Rejecting Null Hypothesis

- ▶ The usual F -test rejects H_0 at level α is
$$F = \frac{MST}{MSE} > F_{k-1, n-k}(\alpha)$$
- ▶ Equivalently, reject H_0 for large values of $\frac{SST}{SSE}$ or for large values of $1 + \frac{SST}{SSE}$
- ▶ The statistic appropriate for a multivariate generalization rejects H_0 for small values of

$$\frac{1}{1 + \frac{SST}{SSE}} = \frac{SSE}{SST + SSE}$$

Generalizing to Multivariate Data

- ▶ Generalizing multivariate T^2 to more than two populations.
- ▶ Suppose that we have random samples from k populations and measures on p variables:

Population 1: $X_{11}, X_{12}, \dots, X_{1n_1}$

Population 2: $X_{21}, X_{22}, \dots, X_{2n_2}$

$\vdots$

Population k : $X_{k1}, X_{k2}, \dots, X_{kn_k}$

where each x_{lj} is a $p \times 1$ vector.

ANOVA- One Way: Data Table

		Treatment			
		1	2	...	k
Subject	1	X_{11}	X_{21}	...	X_{k1}
	2	X_{12}	X_{22}		X_{k2}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	n_i	X_{1n_1}	X_{2n_2}	...	X_{kn_k}

Example

Consider the following independent samples from three populations

Population 1	Population 2	Population 3
9	0	3
6	2	1
9		2

MANOVA- One Way: Data Table

		Treatment			
		1	2		k
Subject	1	$X_{11} = \begin{bmatrix} X_{111} \\ X_{112} \\ \vdots \\ X_{11p} \end{bmatrix}$	$X_{21} = \begin{bmatrix} X_{211} \\ X_{212} \\ \vdots \\ X_{21p} \end{bmatrix}$	$\dots$	$X_{k1} = \begin{bmatrix} X_{k11} \\ X_{k12} \\ \vdots \\ X_{k1p} \end{bmatrix}$
	2	$X_{12} = \begin{bmatrix} X_{121} \\ X_{122} \\ \vdots \\ X_{12p} \end{bmatrix}$	$X_{22} = \begin{bmatrix} X_{221} \\ X_{222} \\ \vdots \\ X_{22p} \end{bmatrix}$		$X_{k2} = \begin{bmatrix} X_{k21} \\ X_{k22} \\ \vdots \\ X_{k2p} \end{bmatrix}$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	n_i	$X_{1n_i} = \begin{bmatrix} X_{1n_i1} \\ X_{1n_i2} \\ \vdots \\ X_{1n_ip} \end{bmatrix}$	$X_{2n_i} = \begin{bmatrix} X_{2n_i1} \\ X_{2n_i2} \\ \vdots \\ X_{2n_ip} \end{bmatrix}$	$\dots$	$X_{kn_i} = \begin{bmatrix} X_{kn_i1} \\ X_{kn_i2} \\ \vdots \\ X_{kn_ip} \end{bmatrix}$

X_{ijk} : Observation for variable k from subject j in treatment i .

ANOVA vs MANOVA Data Structure

One Way ANOVA (p = 1 variable)

Subject	Treatment (Population)		
	1	2	3
1	9	0	3
2	6	2	1
3	9		2

$$X_{ij} \in \mathbb{R}$$

X_{ij} = single measurement

Compare:

$$\mu_1, \mu_2, \mu_3$$

ANOVA compares **means** vs

One Way MANOVA (p = 2 variables)

Subject	Treatment (Population)		
	1	2	3
1	$\begin{bmatrix} 9 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 3 \\ 8 \end{bmatrix}$
2	$\begin{bmatrix} 6 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 9 \end{bmatrix}$
3	$\begin{bmatrix} 9 \\ 7 \end{bmatrix}$		$\begin{bmatrix} 2 \\ 7 \end{bmatrix}$

$$X_{ij} = \begin{bmatrix} X_{ij1} \\ X_{ij2} \end{bmatrix} \in \mathbb{R}^2$$

Compare: μ_1, μ_2, μ_3

MANOVA compares **mean vectors**

Basic Assumptions needed for Statistical Inference

- ▶ $X_{i1}, X_{i2}, \dots, X_{in_i}$ is a random sample of size n_i from a population with means μ_i for $i = 1, 2, \dots, k$ (i.e., **observations within populations are independent** and representative of their populations).
- ▶ Random samples from **different populations are independent**.
- ▶ All populations have the **same covariance matrix Σ** .
- ▶ $X_{ij} \sim N(\mu_i, \Sigma)$; that is, **each population is multivariate normal**.

One-Way MANOVA

- MANOVA model for comparing k population mean vectors parallels univariate ANOVA:

$$\begin{array}{ccccccc}
 X_{lj} & = & \mu & + & \tau_l & + & \epsilon_{lj} \\
 \left(\begin{array}{c} \text{observation} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} \text{overall} \\ \text{mean} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} l^{\text{th}} \text{ trt} \\ \text{effect} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} \text{residual} \\ j^{\text{th}} \text{ group} \\ j^{\text{th}} \text{ case} \end{array} \right) \\
 \underbrace{p \times 1}_{\text{Random}} & & \underbrace{p \times 1}_{\text{Fixed}} & & \underbrace{p \times 1}_{\text{Fixed}} & & \underbrace{p \times 1}_{\text{Random}}
 \end{array}$$

where $\epsilon_{lj} \sim N_p(0, \Sigma)$ and all independent.

- For identification $\sum_{l=1}^k n_l \tau_l = 0$.

Observation Vectors

- ▶ Each component of X_{lj} satisfies the 1-way ANOVA model.
- ▶ The errors for the components of X_{lj} are correlated, but the covariance matrix Σ is same for all populations.

- ▶ The observation vector X_{lj} is decomposed as

$$\begin{array}{ccccccc}
 x_{lj} & = & \bar{x} & + & (\bar{x}_l - \bar{x}) & + & (x_{lj} - \bar{x}_l) \\
 \left(\begin{array}{c} \text{observation} \end{array} \right) & & \left(\begin{array}{c} \text{overall} \\ \text{sample} \\ \text{mean} \end{array} \right) & & \left(\begin{array}{c} l^{th} \text{ est.} \\ \text{trt} \\ \text{effect} \end{array} \right) & & \left(\begin{array}{c} \text{residual} \end{array} \right) \\
 & = & \hat{\mu} & + & \hat{\tau}_l & + & \hat{\epsilon}_{lj}
 \end{array}$$

- ▶ We also have a decomposition of sums-of-squares and crossproducts, or “SSCP” for short.

Sums-of-Squares and Cross Products

- The total corrected squares and cross-products

$$\begin{aligned}(x_{lj} - \bar{x})(x_{lj} - \bar{x})' &= [(x_{lj} - \bar{x}_l) + (\bar{x}_l - \bar{x})][(x_{lj} - \bar{x}_l) + (\bar{x}_l - \bar{x})]' \\ &= \underbrace{(x_{lj} - \bar{x}_l)(x_{lj} - \bar{x}_l)' + (\bar{x}_l - \bar{x})(\bar{x}_l - \bar{x})'}_{\text{squares \& cross products}}\end{aligned}$$

squares & cross products

$$\underbrace{(x_{lj} - \bar{x}_l)(\bar{x}_l - \bar{x})' + (\bar{x}_l - \bar{x})(x_{lj} - \bar{x}_l)'}_{\text{cross- products}}$$

Sum of Squares

$$\begin{aligned}
 \sum_{j=1}^{n_l} (x_{lj} - \bar{x})(x_l - \bar{x})' &= \left(\sum_{j=1}^{n_l} (x_{lj} - \bar{x}_l) \right) (\bar{x}_l - \bar{x})' \\
 &= \left(\left(\sum_{j=1}^{n_l} x_{lj} \right) - n_l \bar{x}_l \right) (\bar{x}_l - \bar{x})' \\
 &= n_l (\bar{x}_l - \bar{x}_l) (\bar{x}_l - \bar{x})' = 0
 \end{aligned}$$

Summing the rest over j and l we get

$$\begin{aligned}
 \sum_{l=1}^k \sum_{j=1}^{n_l} (x_{lj} - \bar{x})(x_{lj} - \bar{x})' &= \sum_{l=1}^k n_l (\bar{x}_l - \bar{x})(\bar{x}_l - \bar{x})' + \sum_{l=1}^k \sum_{j=1}^{n_l} (x_{lj} - \bar{x}_l)(x_{lj} - \bar{x}_l)' \\
 \text{Total(corrected)SSCP} &= \text{Treatment} + \text{Residual} \\
 &= \text{Between Groups} + \text{Within Groups}
 \end{aligned}$$

Within Groups SSCP

$$\begin{aligned} W(\text{or } E) &= \sum_{l=1}^k \sum_{j=1}^{n_l} (x_{lj} - \bar{x}_l)(x_{lj} - \bar{x}_l)' \\ &= \sum_{j=1}^{n_1} (x_{1j} - \bar{x}_1)(x_{1j} - \bar{x}_1)' + \dots + \sum_{j=1}^{n_k} (x_{kj} - \bar{x}_k)(x_{kj} - \bar{x}_k)' \\ &= (n_1 - 1)S_1 + \dots + (n_k - 1)S_k \end{aligned}$$

W is proportional to a pooled estimated of the common Σ .

Between Groups SSCP and Test Statistic

- ▶ With respect to between groups SSCP,

$$B(\text{ or } H) = \sum_{l=1}^k n_l (\bar{x}_l - \bar{x})(\bar{x}_l - \bar{x})' = \sum_{l=1}^k n_l \hat{\tau}_l \hat{\tau}_l'$$

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- ▶ To test H_0 , we consider the ratio of generalized SSCP's

$$\Lambda^* = \frac{|W|}{|W + H|} = \frac{|W|}{|T|}$$

where $T = W + H$ is the total corrected SSCP.

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- ▶ Λ^* is known as “Wilk's Lambda”

Hypothesis Testing with Λ^*



$$\Lambda^* = \frac{|W|}{|T|} = \frac{\prod_{i=1}^p \lambda_i}{\prod_{i=1}^p \lambda_i^*}$$

where λ_i are eigen values of W and λ_i^* are eigen values of T .

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- ▶ $H_0 : \tau_1 = \tau_2 = \dots = \tau_k = 0$ is **false** then H is **not** close to 0.
This implies that values on diagonals of T will be large which further implies $\lambda_i < \lambda_i^*$. This implies Λ^* is “small”.

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This implies that values on diagonals of T will be large which further implies $\lambda_i < \lambda_i^*$. This implies Λ^* is “small”.
- The exact distribution of Λ^* can be derived for special cases of p and k .

Distribution of Wilks' Lambda

No. of vari-ables	No. of groups	Sampling distribution for multivariate normal data
$p = 1$	$g \geq 2$	$\left(\frac{\sum n_t - g}{g - 1} \right) \left(\frac{1 - \Lambda^*}{\Lambda^*} \right) \sim F_{g-1, \sum n_t - g}$
$p = 2$	$g \geq 2$	$\left(\frac{\sum n_t - g - 1}{g - 1} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) \sim F_{2(g-1), 2(\sum n_t - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\frac{\sum n_t - p - 1}{p} \right) \left(\frac{1 - \Lambda^*}{\Lambda^*} \right) \sim F_{p, \sum n_t - p - 1}$
$p \geq 1$	$g = 3$	$\left(\frac{\sum n_t - p - 2}{p} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) \sim F_{2p, 2(\sum n_t - p - 2)}$

Other Test Statistics

1. Wilk's Lambda: $\Lambda^* = \frac{|W|}{|W+H|}$

2. Hotelling-Lawley Trace:

$$\text{trace}(W^{-1}H) = \text{trace}(HW^{-1}) = \sum_{i=1}^k \tilde{\lambda}_i, \text{ where } \tilde{\lambda}_i \text{ is eigen value of } HE^{-1}$$

▶ Reject H_o when $\text{trace}(HW^{-1})$ is large.

▶ When H_o is true, $\text{tr}(HW^{-1}) \sim \chi^2_{p(k-1)}$

3. Pillai Trace: $\text{trace}(H(H+W)^{-1}) = \sum_{i=1}^p \frac{\tilde{\lambda}_i}{1 + \tilde{\lambda}_i}$

▶ Reject the null hypothesis if this test statistic is large.

4. Roy's Maximum Root: Largest value of HW^{-1} .

▶ Reject the null hypothesis if this test statistic is large.

Which Test Statistic to Use

- ▶ If all statistics lead to the same conclusion, use Λ^* .
- ▶ If statistics lead to different conclusion, need to figure out why.
- ▶ From the simulations reported to date, the first three tests have similar power, while the last, Roy's test, behaves differently.
 - ▶ Its power is best only when there is a single nonzero eigenvalue and, at the same time, the power is large-one treatment is much more different from the rest.
- ▶ Pillai's trace is slightly more robust against nonnormality-least affected against departures from normality

One Sample Mean Test

Goal: Compare a sample mean with a known population value.

One Sample Mean Test

Univariate Hypothesis

$$H_0 : \mu = \mu_0$$

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Test Statistic (t-test)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$t \sim t_{n-1}$$

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Multivariate Extension (Hotelling T^2)

Now each observation is a vector

$$\mathbf{x} = (x_1, x_2, \dots, x_p)'$$

$$H_0 : \mu = \mu_0$$

$$T^2 = n(\bar{\mathbf{x}} - \mu_0)' S^{-1} (\bar{\mathbf{x}} - \mu_0)$$

$$\frac{n-p}{p(n-1)} T^2 \sim F_{p, n-p}$$

Two Sample Mean Comparison

Goal: Compare the means of two populations.

Two Sample Mean Comparison

Univariate Hypothesis

$$H_0 : \mu_1 = \mu_2$$

Two Sample Mean Comparison

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Pooled Variance

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

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Two-sample t-test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$t \sim t_{n_1 + n_2 - 2}$$

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Multivariate Two Sample (Hotelling T^2)

$$H_0 : \mu_1 = \mu_2$$

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)' S_p^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$$

$$\frac{(n_1 + n_2 - p - 1)}{p(n_1 + n_2 - 2)} T^2 \sim F_{p, n_1 + n_2 - p - 1}$$

ANOVA and MANOVA

Problem: Compare more than two groups.

ANOVA and MANOVA

One-Way ANOVA Hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_g$$

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Variance Decomposition

$$SS_T = SS_B + SS_W$$

(Total = Between + Within variation)

ANOVA and MANOVA

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ANOVA Test Statistic

$$F = \frac{MS_B}{MS_W}$$

ANOVA and MANOVA

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Multivariate Case

Each observation is a vector:

$$\mathbf{x} = (x_1, x_2, \dots, x_p)'$$

We use matrices:

$$W = \text{Within SSCP}, \quad B = \text{Between SSCP}$$

ANOVA and MANOVA

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MANOVA Hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_g$$

$$\text{Wilks' Lambda } \Lambda = \frac{|W|}{|B + W|}$$

Independent Samples and Blocks

- ▶ A **randomized block design** experiment reduces the variation **within** the samples, making it easier to detect differences between populations.

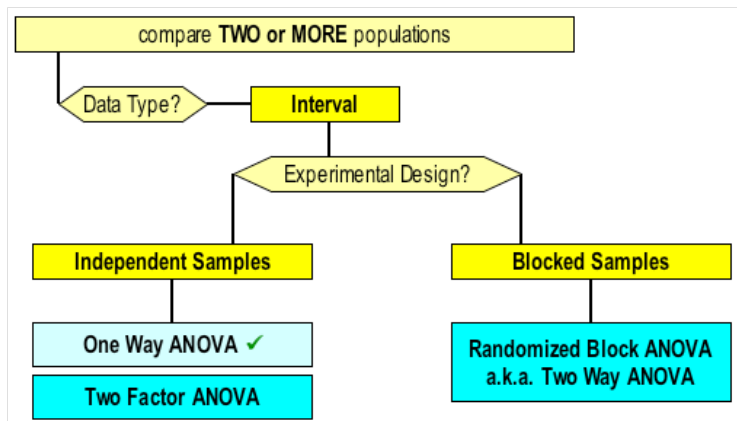
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Independent Samples and Blocks

- ▶ A **randomized block design** experiment reduces the variation **within** the samples, making it easier to detect differences between populations.
- ▶ The term **block** refers to a matched group of observations from each population.
- ▶ We can also perform a blocked experiment by using the same subject for each treatment in a “**repeated measures**” experiment.

- ▶ The randomized block experiment is also called the two-way analysis of variance, not to be confused with the two-factor analysis of variance. To illustrate where we're headed



Randomized Block Analysis of Variance

- ▶ The purpose of designing a randomized block experiment is to reduce the within-treatments variation to more easily detect differences between the treatment means.

Randomized Block Analysis of Variance

- ▶ The purpose of designing a randomized block experiment is to reduce the within-treatments variation to more easily detect differences between the treatment means.
- ▶ In this design, we partition the total variation into three sources of variation:

$$SS(Total) = SST + SSB + SSE$$

where SSB , the **sum of squares for blocks**, measures the variation between the blocks.

RBD: Data Table

In addition to k treatments, we introduce notation for b blocks in our experimental design

		Treatments			
		1	2	...	k
Block	1	X_{11}	X_{21}	...	X_{k1}
	2	X_{12}	X_{22}		X_{k2}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	b	X_{1b}	X_{2b}	...	X_{kb}

Design

		Treatments				Mean
		1	2	...	k	
Block	1	X_{11}	X_{21}	...	X_{k1}	$\bar{X}_{.1}$
	2	X_{12}	X_{22}		X_{k2}	$\bar{X}_{.2}$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	b	X_{1b}	X_{2b}	...	X_{kb}	$\bar{X}_{.b}$
		$\bar{X}_{1.}$	$\bar{X}_{2.}$	...	$\bar{X}_{k.}$	$\bar{X}$

The Sum of Squares: Randomized Block

► Sum of Squared Observations

$$SS_{obs} = SS_{total} = \sum_{l=1}^k \sum_{j=1}^b (X_{lj} - \bar{X})^2$$

► The components of X_{lj} can be taken and sum of squares can be formed as

$$\text{► } SS_{treatment} = \sum_{l=1}^k \sum_{j=1}^b \hat{\tau}_l^2 = \sum_{l=1}^k \sum_{j=1}^b (\bar{X}_{.l} - \bar{X})^2 = \sum_{l=1}^k b(\bar{X}_{.l} - \bar{X})^2$$

$$\text{► } SS_{blocks} = \sum_{l=1}^k \sum_{j=1}^b \hat{\beta}_j^2 = \sum_{l=1}^k \sum_{j=1}^b (\bar{X}_{.j} - \bar{X})^2 = \sum_{j=1}^b k(\bar{X}_{.j} - \bar{X})^2$$

$$\text{► } SS_{residual} = \sum_{l=1}^k \sum_{j=1}^b \hat{\epsilon}_{ij}^2 = \sum_{l=1}^k \sum_{j=1}^b (X_{lj} - \bar{X}_{.l} - \bar{X}_{.j} + \bar{X})^2$$

Anova Table

We can summarize this new information in an analysis of variance (ANOVA) table for the randomized block analysis of variance as follows

Source of Variation	d.f.:	Sum of Squares	Mean Square	<i>F</i> Statistic
Treatments	$k-1$	SST	$MST = SST/(k-1)$	$F = MST/MSE$
Blocks	$b-1$	SSB	$MSB = SSB/(b-1)$	$F = MSB/MSE$
Error	$n-k-b+1$	SSE	$MSE = SSE/(n-k-b+1)$	
Total	$n-1$	SS(Total)		

Developing an Understanding of Statistical Concepts

- ▶ The randomized block experiment is an extension of the matched pairs experiment.
- ▶ In the matched pairs experiment, we simply remove the effect of the variation caused by differences between the experimental units. The effect of this removal is seen in the decrease in the value of the standard error and the increase in the value of the t -statistic.

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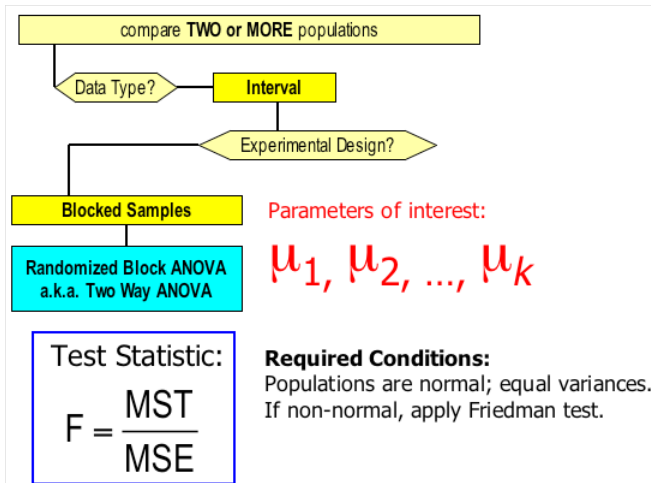
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- ▶ In the randomized block experiment of the analysis of variance, we actually measure the variation between the blocks by computing SSB.
- ▶ The sum of squares for error is reduced by SSB, making it easier to detect differences between the treatments.
- ▶ Additionally, we can test to determine whether the blocks differ—a procedure we were unable to perform in the matched pairs experiment.

Summary:RBD

Factors that Identify the Randomized Block of the Analysis of Variance



Outline

One-Way Analysis of Variance

One-Way Manova

Two Way Analysis of Variance

Two Way MANOVA

Two-Factor Multivariate Analysis of Variance

Univariate Two-Way Fixed Effects Model

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	b	X_{1b}	X_{2b}	...	X_{kb}

MANOVA RBD: Data Table

		Treatments			
		1	2	...	k
Block	1	$X_{11} = \begin{bmatrix} X_{111} \\ X_{112} \\ \vdots \\ X_{11p} \end{bmatrix}$	$X_{21} = \begin{bmatrix} X_{211} \\ X_{212} \\ \vdots \\ X_{21p} \end{bmatrix}$	...	$X_{k1} = \begin{bmatrix} X_{k11} \\ X_{k12} \\ \vdots \\ X_{k1p} \end{bmatrix}$
	2	$X_{12} = \begin{bmatrix} X_{121} \\ X_{122} \\ \vdots \\ X_{12p} \end{bmatrix}$	$X_{22} = \begin{bmatrix} X_{221} \\ X_{222} \\ \vdots \\ X_{22p} \end{bmatrix}$		$X_{k2} = \begin{bmatrix} X_{k21} \\ X_{k22} \\ \vdots \\ X_{k2p} \end{bmatrix}$
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	b	$X_{1b} = \begin{bmatrix} X_{1b1} \\ X_{1b2} \\ \vdots \\ X_{1bp} \end{bmatrix}$	$X_{2b} = \begin{bmatrix} X_{2b1} \\ X_{2b2} \\ \vdots \\ X_{2bp} \end{bmatrix}$	...	$X_{kb} = \begin{bmatrix} X_{kb1} \\ X_{kb2} \\ \vdots \\ X_{kbp} \end{bmatrix}$

The Additive Model

$$\begin{array}{ccccccc}
 X_{lj} & = & \mu & + & \tau_l & + & \beta_j & + & \epsilon_{lj} \\
 \left(\begin{array}{c} \text{observation} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} \text{overall} \\ \text{mean} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} j^{\text{th}} \text{ trt} \\ \text{effect} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} j^{\text{th}} \text{ block} \\ \text{effect} \\ \text{vector} \end{array} \right) & & \left(\begin{array}{c} \text{residual} \\ j^{\text{th}} \text{ trt} \\ j^{\text{th}} \text{ block} \end{array} \right) \\
 \underbrace{p \times 1} & & \underbrace{p \times 1} & & \underbrace{p \times 1} & & \underbrace{p \times 1} & & \underbrace{p \times 1} \\
 \text{Random} & & \text{Fixed} & & \text{Fixed} & & \text{Fixed} & & \text{Random}
 \end{array}$$

In this model :

- ▶ X_{lj} is the $p \times 1$ vector of observations for treatment l in block j ; and is written as a function of the following
 - ▶ μ_k is the overall mean for variable k ; these are collected into the overall mean vector μ
 - ▶ τ_{lk} is the effect of treatment l on variable k ; these are collected into the treatment effect vector τ_l
 - ▶ β_{jk} is the effect of block j on variable k ; these are collected in the block effect vector β_j
 - ▶ ϵ_{ljk} is the experimental error for treatment l , block j , and variable k ; these are collected into the error vector ϵ_{lj}

Assumptions

- ▶ The error vectors ϵ_{lj} have zero population mean;
- ▶ The error vectors ϵ_{lj} have common variance-covariance matrix Σ
- ▶ The error vectors ϵ_{lj} are independently sampled from a multivariate normal distribution;
- ▶ There is no block by treatment interaction. This means that the effect of the treatment is not affected by, or does not depend on the block.

SSCP Matrices

In this case the total sum of squares and cross products matrix may be partitioned into three matrices, three different sum of squares cross product matrices:

$$T = H + B + W$$

- ▶ H is the Treatment Sum of Squares and Cross Products matrix;
- ▶ B is the Block Sum of Squares and Cross Products matrix;
- ▶ W is the Error Sum of Squares and Cross Products matrix.

MANOVA Table

Source	d.f	SS
Blocks	$b - 1$	B
Treatment	$k - 1$	H
Error	$(b - 1)(k - 1)$	W
Total	$kb - 1$	T

Hypothesis Tests

- ▶ To test the null hypothesis that the treatment mean vectors are equal, compute a Wilks' Lambda using the following expression:

$$\Lambda^* = \frac{|W|}{|W + H|}$$

- ▶ To test the null hypothesis that there are no block effects, compute a Wilks' Lambda using the following expression:

$$\Lambda^* = \frac{|W|}{|W + B|}$$

- └ Two-Factor Multivariate Analysis of Variance
 - └ Univariate Two-Way Fixed Effects Model

Outline

One-Way Analysis of Variance

One-Way Manova

Two Way Analysis of Variance

Two Way MANOVA

Two-Factor Multivariate Analysis of Variance

Univariate Two-Way Fixed Effects Model

Two Factor Design

- ▶ A multifactor experiment is one where there are two or more factors that define the treatments.
- ▶ For example, if instead of just varying the advertising strategy for a new apple juice product we also varied the advertising medium (e.g. television or newspaper), then we have a two-factor analysis of variance situation.
- ▶ The first factor, advertising strategy, has three levels (convenience, quality, and price) while the second factor, advertising medium, has two levels (TV or print).

Factorial Experiment

- ▶ We address the problem where the experiment features two factors.
- ▶ The general term for such data-gathering procedures is factorial experiment.
- ▶ In factorial experiments, we can examine the effect on the response variable of two or more factors
- ▶ We can use the analysis of variance to determine whether the Levels of each factor are different from one another.

Factorial Experiment

- ▶ A complete **factorial experiment** is an experiment in which the data for all possible combinations of the levels of the factors are gathered. This is also known as a two-way classification.
- ▶ The two factors are usually labeled A & B , with the number of levels of each factor denoted by a & b respectively.
- ▶ The number of observations for each combination is called a replicate, and is denoted by n . For our purposes, the number of replicates will be the same for each treatment, that is they are **balanced**.

Example: Factorial Design

Less than high school

Male

Female

10	7
9	13
12	14
16	6
14	11
17	14
13	13
9	11
11	14
15	12
12	7
11	12
9	6
14	15
12	10
16	13
10	9
10	15
5	12
11	13
15	5
8	13
7	12
7	3
7	13
9	11
14	15
15	5
11	9
13	8
8	7
9	9
5	3
11	7
13	9
8	6
7	10
11	15
10	4
8	11

High School

Less than Bachelor's degree

At least one Bachelor's degree

ANOVA Table

Source of Variation	d.f.:	Sum of Squares	Mean Square	<i>F</i> Stat
Factor A	$a-1$	$SS(A)$	$MS(A)=SS(A)/(a-1)$	$F=MS(A)$
Factor B	$b-1$	$SS(B)$	$MS(B)=SS(B)/(b-1)$	$F=MS(B)$
Interaction	$(a-1)(b-1)$	$SS(AB)$	$MS(AB) = \frac{SS(AB)}{[(a-1)(b-1)]}$	$F=MS(AB)$
Error	$ab(n-1)$	SSE	$MSE=SSE/(ab(n-1))$	
Total	$nab-1$	$SS(Total)$		

Hypothesis

- ▶ Test for the differences between the Levels of Factor A ?
 - ▶ H_0 : The means of the a levels of Factor A are equal
 - ▶ H_1 : At least two means differ
 - ▶ Test statistic: $F = MS(A)/MSE$
- ▶ Test for the differences between the Levels of Factor B ?
 - ▶ H_0 : The means of the a levels of Factor B are equal
 - ▶ H_1 : At least two means differ
 - ▶ Test statistic: $F = MS(B)/MSE$
- ▶ Test for interaction between Factors A and B ?
 - ▶ H_0 : Factors A and B do not interact to affect the mean responses.
 - ▶ H_1 : Factors A and B do interact to affect the mean responses.
 - ▶ Test statistic: $F = MS(AB)/MSE$

Univariate Model with Interaction

- ▶ Let X_{lkr} denote the r^{th} observation at level l of factor A and level k of factor B .
- ▶ We can specify the univariate two-way model as

$$X_{lkr} = \mu + \tau_l + \beta_k + \gamma_{lk} + \epsilon_{lkr}$$

- ▶ $l = 1, 2, \dots, a$
- ▶ $k = 1, 2, \dots, b$.
- ▶ $r = 1, 2, \dots, n$

where $\sum_{l=1}^a \tau_l = \sum_{k=1}^b \beta_k = \sum_{l=1}^a \gamma_{lk} = \sum_{k=1}^b \gamma_{lk} = 0$ and ϵ_{lkr} are independent $N(0, \sigma^2)$ random variables.

Expected Response

The expected response at l^{th} level of factor A and k^{th} level of factor B is thus

$$\begin{aligned}
 E(X_{\ell k}) &= \mu + \tau_{\ell} + \beta_k + \gamma_{\ell k} \\
 \left(\begin{array}{c} \text{mean} \\ \text{response} \end{array} \right) &= \left(\begin{array}{c} \text{overall} \\ \text{level} \end{array} \right) + \left(\begin{array}{c} \text{effect of} \\ \text{factor 1} \end{array} \right) + \left(\begin{array}{c} \text{effect of} \\ \text{factor 2} \end{array} \right) + \left(\begin{array}{c} \text{factor 1-factor 2} \\ \text{interaction} \end{array} \right)
 \end{aligned}$$

The presence of interaction, $\gamma_{\ell k}$, implies that the factor effects are not additive and complicates the interpretation of effects.

ANOVA and t -tests of 2 means

- ▶ Why do we need the analysis of variance? Why not test every pair of means? For example say $k = 6$.
- ▶ There are $6_{C_2} = 6(5)/2 = 15$ different pairs of means.
 - ▶ 1&2 1&3 1&4 1&5 1&6
 - ▶ 2&3 2&4 2&5 2&6
 - ▶ 3&4 3&5 3&6
 - ▶ 4&5 4&6
 - ▶ 5&6

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- ▶ If we test each pair with $\alpha = .05$ we increase the probability of making a Type I error. If there are no differences then the probability of making at least one Type I error is $1 - (.95)^{15} = 1 - .463 = .537$

ANOVA versus t -tests

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ANOVA versus t -tests

- ▶ We have argued that the t -tests cannot replace analysis of variance
- ▶ Can we use the analysis of variance instead of t -tests?
 - ▶ Anova allows us to test for a difference only.
 - ▶ Anova requires equal population variances.

Checking the Required Conditions

- ▶ The F -test of the analysis of variance requires that the random variable be normally distributed with **equal variances**. The normality requirement is easily checked graphically by producing the histograms for each sample.

Checking the Required Conditions

- ▶ The F -test of the analysis of variance requires that the random variable be normally distributed with **equal variances**. The normality requirement is easily checked graphically by producing the histograms for each sample.
- ▶ The equality of variances is examined by printing the sample standard deviations or variances. The similarity of sample variances allows us to assume that the population variances are equal.

Violation of the Required Conditions

- ▶ If the data are not normally distributed we can replace the one-way analysis of variance with its nonparametric counterpart, which is the Kruskal-Wallis test.

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Violation of the Required Conditions

- ▶ If the data are not normally distributed we can replace the one-way analysis of variance with its nonparametric counterpart, which is the Kruskal-Wallis test.
- ▶ If the population variances are unequal, we can use several methods to correct the problem.
- ▶ However, these corrective measures are beyond the level of the course.